UNIT - I
Introduction to Automation

Automation and Robotics (23A03606) - handwritten study notes

1. Introduction to Automation

Definition

- **Automation** is the technology by which a process or procedure is performed **without human assistance**, using programmed commands combined with automatic feedback control.
- The word comes from the Greek **automatos** = self-acting. The term was coined by D.S. Harder (Ford Motor Company, 1946).
- A programmed set of instructions + a control system + a power source = an automated system.

> Automation = Power source + Program of instructions + Control system that actuates the instructions.

Need / Reasons for Automation

- Increase **labour productivity** (more output per man-hour)
- Reduce labour cost and shortage of skilled labour
- Mitigate the effects of labour shortage
- Reduce or eliminate routine manual and clerical tasks
- Improve worker **safety** (hazardous, hot, toxic work)
- Improve **product quality** and uniformity
- Reduce manufacturing lead time and work-in-process
- Accomplish processes that cannot be done manually (micro-electronics, high precision)
- Avoid the high cost of not automating (competition)

Advantages and Disadvantages

Advantages	Disadvantages
Higher production rate and productivity	High initial investment
Consistent quality, less scrap	Needs skilled maintenance staff
Lower unit cost in the long run	Unemployment / worker resistance
Safer working conditions	Less flexible for small batches (fixed automation)
Reduced lead time and inventory	Breakdown stops the whole line

Types of Automation

Automated production systems are classified into **three** types:

- **Fixed (hard) automation**
  - Sequence of operations fixed by equipment configuration; changing the product needs new equipment.
  - High initial cost, very high production rate, low unit cost. Suited to **mass production**.
  - Examples: transfer lines, automatic assembly machines, bottling plants.
- **Programmable automation**
  - Equipment can be reprogrammed for a new product; changeover time is spent in reprogramming and setup.
  - Suited to **batch production** with medium volume and variety. Examples: NC machines, industrial robots, PLCs.
- **Flexible (soft) automation**
  - An extension of programmable automation: changeover is almost **zero lost time**, different parts mixed on the same system.
  - Examples: Flexible Manufacturing System (FMS), CNC machines with automatic tool and pallet change.

Feature	Fixed	Programmable	Flexible
Volume	Very high	Medium	Medium
Variety	Very low	Medium-high	High

Investment	High (special)	Medium	Very high
Changeover	Costly	Reprogramming time	Almost none
Production rate	Highest	Medium	Medium

Remember: Fixed ⟶ high volume / low variety. Programmable ⟶ batch. Flexible ⟶ mixed models, no changeover loss.

2. Basic Elements of an Automated System

Every automated system has **three basic elements**:

Power to accomplish the process → Program of instructions → Control system

1. Power to Accomplish the Automated Process

- Electric power is the most common (lighting, motors, controls, computers)
- Also pneumatic and hydraulic power; solar / battery in special cases
- Power is used for the **process** (drive motors, heating) and for **auxiliary functions**: loading/unloading, data processing, communication, material handling.

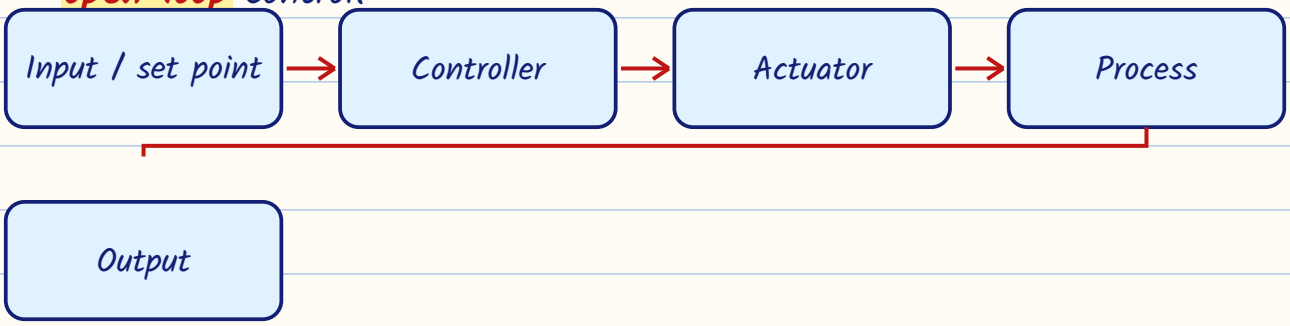
2. Program of Instructions

- The **action program** defines the steps of the process (e.g. a part program for a CNC machine).
- Work cycle program: a sequence of steps for one part (load, clamp, machine, unload).
- Decision-making features: operator interaction, different part styles, variations in raw material.

3. Control System

- Executes the program of instructions: **closed-loop (feedback)** and

open-loop control.



- **Closed loop**: output is measured by a sensor and compared with the reference; the error drives the controller.
- **Open loop**: no feedback; the controller output depends only on the program (e.g. stepper motor drive, washing machine timer).

Open loop	Closed loop
No feedback	Feedback sensor present
Simple and cheap	More complex, costly
Less accurate	Accurate, self-correcting
Stepper motors	Servo motors

3. Advanced Automation Functions

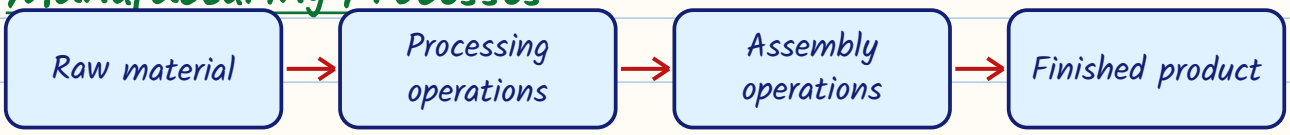
- **Safety monitoring**: detect hazards (fire, collision, overload) and stop or alert.
- **Maintenance and repair diagnostics**: status monitoring, failure diagnosis, preventive maintenance schedule.
- **Error detection and recovery**: error detection, interpretation of the error, and recovery action by the program.

Levels of intelligence: error detection → diagnosis → automatic recovery. Higher the level, fewer human interventions.

4. Manufacturing Industries

- **Manufacturing** = transformation of raw materials into finished products of higher value using physical/chemical processes and machines.
- Industry is classified as:
  - **Primary**: farming, mining, quarrying, oil (use natural resources)
  - **Secondary**: manufacturing of goods, construction, power utilities
  - **Tertiary**: services - banking, retail, education, transport
- Manufacturing industries by product:
  - **Process industries**: chemicals, petroleum, food, paper, cement (continuous or batch processing).
  - **Discrete-item industries**: cars, machines, appliances, electronics (individual parts and assemblies).
  - Products: **consumer goods**, **capital goods** (machine tools, cranes) and **intermediate goods**.

Manufacturing Processes



- **Processing**: shaping (casting, forming, machining), property enhancing (heat treatment), surface processing (cleaning, painting, plating).
- **Assembly**: joining (welding, brazing, soldering), mechanical fastening, adhesive bonding.

5. Types of Production

Production is classified by **quantity** and **product variety**:

Type	Quantity per year	Variety	Layout
Job shop (low)	1 - 100	Very high	Fixed position / process
Batch (medium)	100 - 10,000	Medium	Process or cellular
Mass (high)	10,000 - millions	Low	Product / line layout

- **Job shop production**: one-off or few products, general purpose machines, skilled labour. Eg: prototypes, special machines, ships.
- **Batch production**: lots of the same item produced repeatedly; machines are reset between batches. Eg: books, clothing, machine parts.
  - **Production time of a batch** = setup time + (Q x cycle time).
- **Mass production**: very high volume; two sub types:
  - **Quantity production**: single machine produces one part at a high rate (e.g. stamping).
  - **Flow-line production**: multiple workstations in sequence (e.g. automobile assembly).

Production Quantity vs Variety (P-Q)

- **Product variety**: hard variety (different product types) vs soft variety (small model differences).
- Product variety is inversely related to quantity: more variety ⟶ less quantity per model.

$$\text{Production rate } R_p = 1 / T_c$$

$$\text{Cycle time } T_c = T_o + T_h + T_{th}$$

- T_o = operation time, T_h = handling time, T_{th} = tool handling time.

Production Performance Measures

$$\text{Average production rate } R_p = 60 / T_p$$

$$\text{Availability } A = (MTBF - MTTR) / MTBF$$

$$\text{Utilisation } U = \text{Output} / \text{Capacity}$$

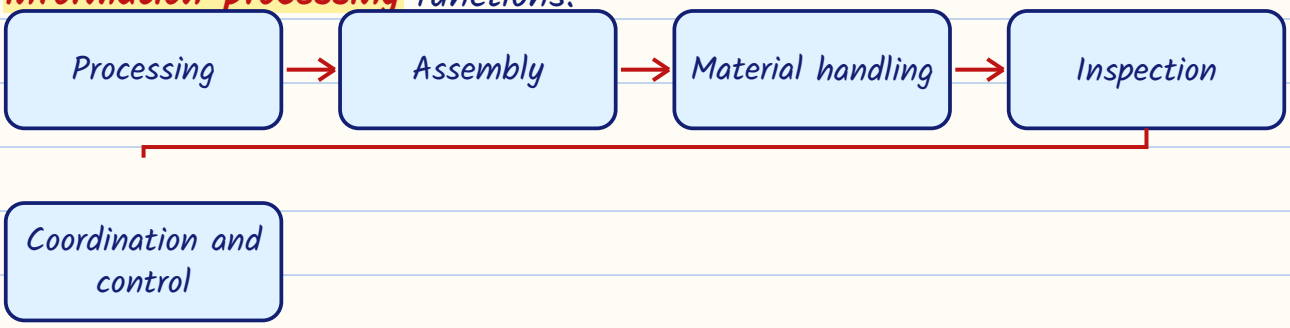
- **Worked example**: T_o = 3 min, T_h = 1 min, T_{th} = 0.5 min ⟶ T_c = 4.5 min ⟶ R_c = 60/4.5 = 13.3 parts/hr.
- If the machine has availability 90%, then actual rate = 13.3 x 0.9 = 12

parts/hr.

MTBF = mean time between failures, MTTR = mean time to repair. Availability falls when repairs take longer.

6. Functions in Manufacturing

Manufacturing has **two groups** of functions: the **physical** and the **information-processing** functions.

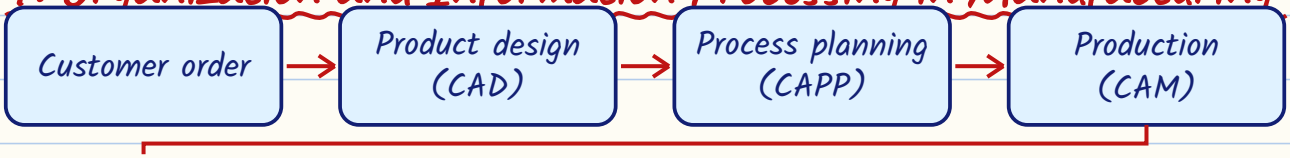


- **Processing and assembly**: the value-adding operations.
- **Material handling and storage**: loading, positioning, transport (AGV, conveyor, forklift), storage (AS/RS).
- **Inspection and test**: quality control, gauging, CMM, vision systems.
- **Coordination and control**: regulation of individual processes (process control) and management of the plant (plant control).

Product Design and Manufacturing Planning

- **Design function**: product concept, CAD, drawings, tolerances.
- **Process planning**: choose operations, machines, tools; route sheet.
- **Production planning**: master schedule, MRP, capacity planning.
- **Quality control**: statistical process control, inspection plans.

7. Organization and Information Processing in Manufacturing



Quality and shipping

- The **information flow** runs in parallel with the physical flow and is managed by computer systems.
- **Product design** ⟶ engineering drawing, bill of materials (BOM).
- **Manufacturing planning**: process planning, master production schedule, MRP, capacity planning.
- **Manufacturing control**: shop-floor control, inventory control, quality control.
- Information processing is organised in **levels**: Enterprise ⟶ Plant ⟶ Cell ⟶ Machine.

CIM: Computer Integrated Manufacturing

- CIM links **CAD, CAM, CAPP, MRP and shop floor control** through a common database and network.
- Benefits: shorter lead time, better quality, lower inventory, faster response.

CAD/CAM = design + manufacturing in one data stream. CIM = CAD/CAM + business functions.

8. Automation Strategies

The **USA principle** (a common sense approach) and the ten strategies for automation:

- **U**nderstand the existing process: input/output, value-added, problems.
- **S**implify the process: eliminate unnecessary steps.
- **A**utomate the simplified process.

The Ten Strategies

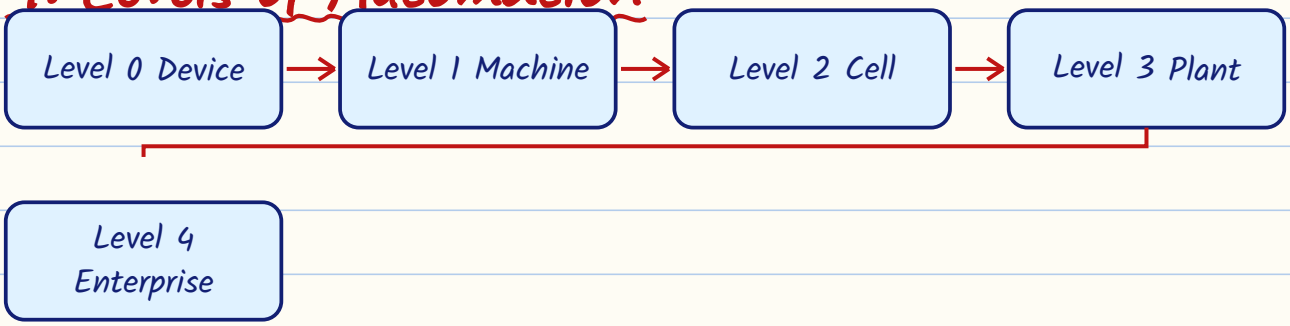
- 1. **Specialization of operations** - dedicated equipment for one task.
- 2. **Combined operations** - several operations on one machine (machining

centre).

- 3. **Simultaneous operations** - perform operations at the same time (multi-spindle drill).
- 4. **Integration of operations** - link workstations with handling, so work moves automatically.
- 5. **Increased flexibility** - quick changeover, programmable machines.
- 6. **Improved material handling and storage** - AGV, AS/RS, conveyors.
- 7. **On-line inspection** - inspect during processing; reduce scrap.
- 8. **Process control and optimisation** - adaptive control, monitor temperature, force etc.
- 9. **Plant operations control** - computer control of inventory, scheduling.
- 10. **Computer integrated manufacturing (CIM)** - integrate all functions.

Mnemonic: S-C-S-I-I-I-O-P-P-C. Learn the keywords of each strategy with one example.

9. Levels of Automation



Level	Name	Examples
0	Device level	Sensors, actuators, motors, valves
1	Machine level	CNC machine, robot, PLC
2	Cell / system level	Manufacturing cell, FMS, assembly line
3	Plant level	Production control, MRP, shop control
4	Enterprise level	Corporate IT, ERP, sales, finance

- As we go up, the **time horizon grows** (milliseconds at device level ⟶ months at enterprise level).
- Lower levels need fast real-time control; higher levels deal with planning data.

10. Hardware Components for Automation and Process Control

Sensors

- Convert physical quantities into electrical signals: **temperature, pressure, flow, position, force, proximity**.
- Types: limit switch, photoelectric, inductive and capacitive proximity, encoders, thermocouples, strain gauges.

Actuators

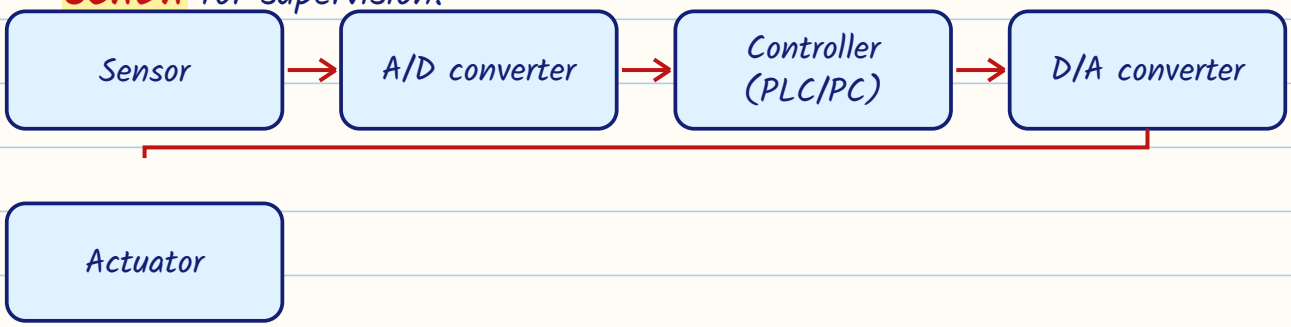
- Convert control signals into physical action: **solenoids, relays, motors (DC, AC, stepper, servo), pneumatic and hydraulic cylinders, valves**.

A/D and D/A Converters

- **ADC**: analogue signals from sensors ⟶ digital values for the controller.
- **DAC**: digital controller output ⟶ analogue voltage for the actuator.

Controllers

- **PLC** (Programmable Logic Controller): rugged, ladder-logic programming, widely used for sequence control.
- **PC based control**, microcontrollers, DCS (distributed control system), **SCADA** for supervision.



Process Control Types

- **Continuous control**: variables vary continuously (temperature, flow) -

PID control.

- **Discrete control**: on/off variables (limit switch, motor on) - PLC and relay logic.
- **Process industries** use continuous + batch; **discrete manufacturing** uses discrete control.

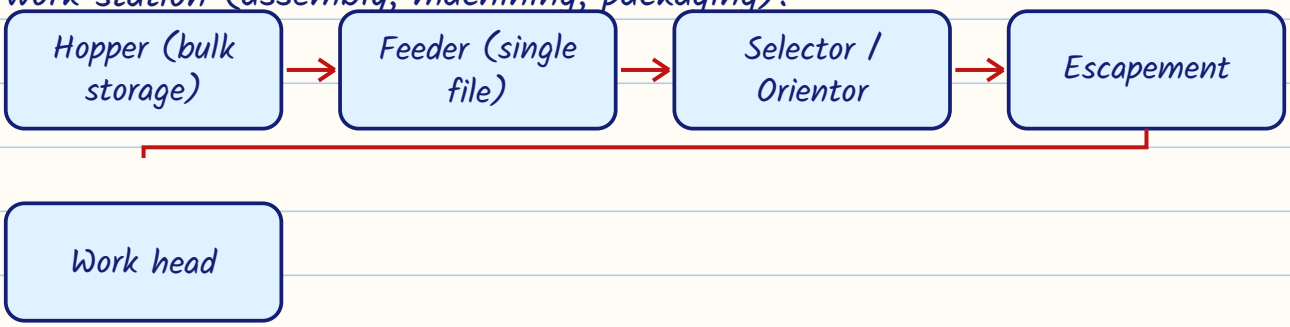
PLC Basics

- Parts: **CPU, input module, output module, power supply, programming device, memory**.
- Programming languages (IEC 61131-3): **ladder diagram**, function block diagram, structured text, instruction list.
- Scan cycle: read inputs ⟶ execute program ⟶ update outputs.

PLC replaces hard-wired relays; easy to modify, small, reliable, with self diagnostics.

11. Mechanical Feeders and Part Handling Devices

Automatic feeding supplies parts at the correct rate and **orientation** to the work station (assembly, machining, packaging).



Hoppers

- A **hopper** is a container that stores a bulk quantity of components in random orientation and feeds them to the feeder.
- Types: **fixed / gravity hoppers**, vibratory hoppers, rotary hoppers.
- Hopper should not jam or damage parts; capacity decides refilling interval.

Feeders

Feeders move parts from the hopper in the correct orientation.

Feeder	Working principle
Vibratory bowl feeder	Vibrating bowl with spiral track; parts climb and are oriented by selectors
Centreboard hopper feeder	Reciprocating centre blade lifts parts into a track
Rotary hopper feeder	Rotating drum / disc picks parts into chute
Magnetic feeder	Magnetic belt/drum for ferrous parts
Belt / roller feeder	Parts carried on conveyor belt
Reciprocating tube feeder	Tube reciprocates in the bulk and picks parts
External gate hopper	Reciprocating gate takes parts from a chute

Vibratory Bowl Feeder (most common)

- A circular bowl with a **helical track** inside; an electromagnet (or piezo) + leaf springs vibrate the bowl at about 50-120 Hz.
- Parts move **up the track with micro-hops**; unwanted orientations are rejected back into the bowl.
- Track tooling (**selectors**) gives the required orientation.
- Advantages: high rates, simple, no moving parts apart from the vibrator, quiet, handles many shapes.
- Limits: noise, not good for delicate or very sticky parts, tooling required for each part.

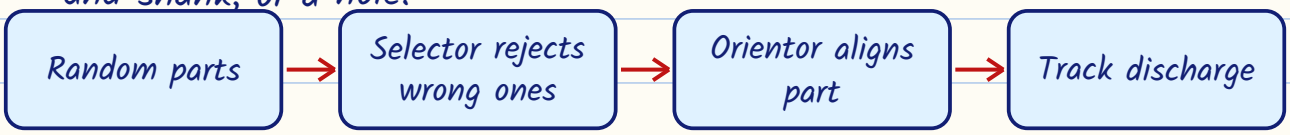
$$\text{Feed rate } f_r = n_p \times v / L$$

- Rate depends on track slope, vibration amplitude and frequency.

Selectors and Orientors

- **Selector**: removes unwanted parts / wrong orientation from the track.
  Examples: **wiper blade, pressure break, rails, notches**.

- **Orientor**: rotates or re-aligns a part into the correct position; works by **gravity and part geometry** using slots, pins, rotating guides.
- Typical orientors: gap (slot) orienting, rotating hook, drop through slot, tilted track.
- Method: use a feature such as an **asymmetric centre of gravity**, a head and shank, or a hole.



- **Example**: a screw / bolt: slot track lets the shank pass while the head rides on rails, so the bolt hangs head-up.
- Techniques: tilt track, wiper blade, pressure break, rail track, stripping gate.

Design rule: use the simplest geometry feature to orient; no complicated mechanism - reliability of a feeder is mostly in its tooling.

Escapements

- Escapement releases **one part at a time** to the work station at the right moment.
- Types: **reciprocating slide, rotary escapement, shuttle, gate type, star wheel**.

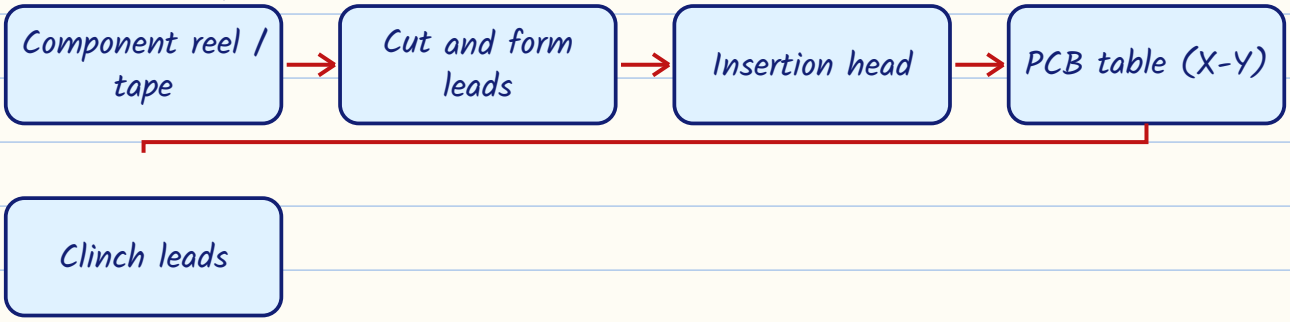
Magazines and Cartridges

- Parts are pre-oriented and stacked in a **magazine** (stack or tube). They need loading in advance but then give very reliable feeding for delicate parts.

Feeding system	Part orientation	Cost	Use
Bowl feeder	Automatic	Medium	Small parts, high rate
Magazine	Pre-oriented	Low	Delicate parts
Flexible feeder	Vision + robot	High	Many part types

12. High-Speed Automatic Insertion Devices

- Used in **electronics manufacturing** (PCB assembly) to insert / place components at high speed.
- Types of insertion:
  - **Radial lead insertion** - for components with two leads on the same side (capacitors, resistors).
  - **Axial lead insertion** - component leads on the same axis (resistors, diodes). The leads are cut and clinched.
  - **DIP insertion** - dual in-line IC packages.
  - **SMD / pick and place** - surface mount devices placed by a vacuum nozzle.



- A **CNC X-Y table** moves the PCB to the correct insertion position.
- Typical speeds: axial inserters 20,000+ components per hour; chip shooters up to 100,000+ cph.
- Features: **tape feeding, programmable sequence, vision alignment, error detection** (missing component).

Other Part Handling Devices

- **Pick-and-place units, parts feeders with robot, conveyors, walking beam, AGVs.**
- **Vacuum cups and magnetic grippers** to pick parts from the track.

Summary of feed system: Hopper (store) ⟶ Feeder (move) ⟶ Selector (reject) ⟶ Orientor (align) ⟶ Escapement (release) ⟶ Work head.

Quick Revision

- Automation: process without human help, power + program + control.
- Types: **fixed**, **programmable**, **flexible**.
- Production types: **job**, **batch**, **mass**.
- Levels of automation: device, machine, cell, plant, enterprise.
- Strategies: specialization, combined, simultaneous, integration, flexibility, handling, inspection, control, plant control, CIM.
- Feeder parts: hopper, feeder, selector, orientor, escapement.

Important Questions

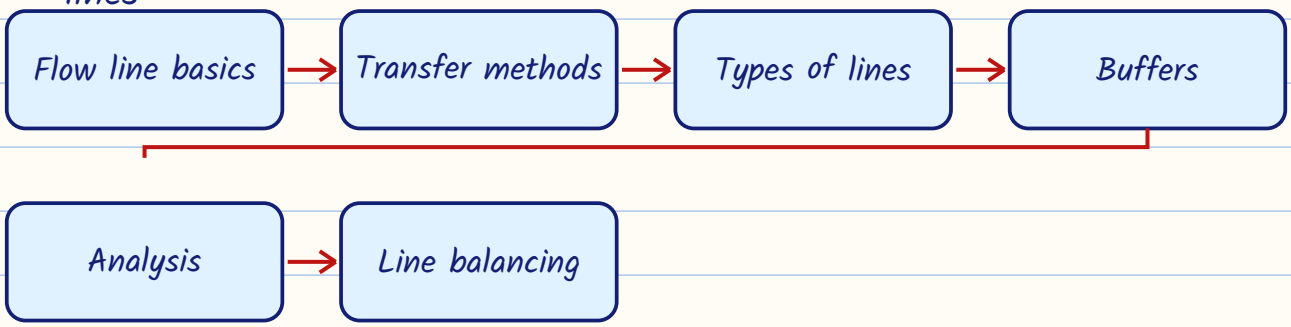
- 1. Define automation. Explain types of automation with examples.
- 2. What are the reasons for automating? Discuss advantages and limitations.
- 3. Explain basic elements of an automated system with a neat sketch.
- 4. Explain types of production and compare them.
- 5. Discuss the ten strategies of automation.
- 6. Explain the levels of automation.
- 7. With a neat sketch explain the vibratory bowl feeder. What are selectors and orientors?
- 8. Write short notes on high-speed automatic insertion devices.

UNIT - II
Automated Flow Lines & Assembly Line Balancing

Automation and Robotics (23A03606) - handwritten study notes

Syllabus at a glance

- Automated flow lines, part transfer methods and mechanisms
- Types of flow lines, flow line with / without buffer storage
- Quantitative analysis of flow lines
- Assembly line balancing: assembly process and systems, assembly line
- Line balancing methods, ways of improving line balance, flexible assembly lines



1. Automated Flow Lines

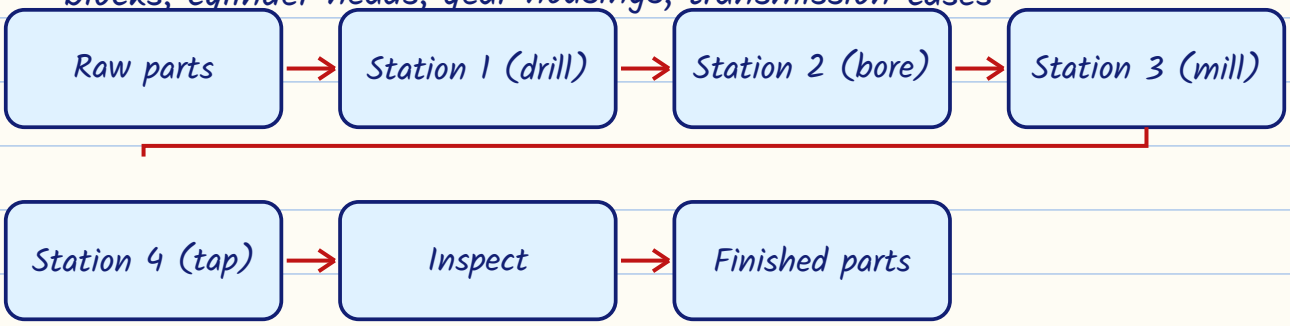
Definition

An **automated flow line** (transfer line) is a series of automatic workstations connected by a **work-part transfer system**. Raw work-parts enter at one end, are processed station by station and leave as finished (or semi-finished) parts at the other end.

- Each station does a **different operation** (drilling, boring, milling, tapping, inspection ...)
- All stations work **simultaneously** on different parts, so output = one part every cycle
- Part movement is **automatic** and stations are **synchronised** by a

common control system

- Used for **mass production** of identical or very similar parts: engine blocks, cylinder heads, gear housings, transmission cases



Disadvantages and Limitations

- Very high capital investment; economic only for large volumes
- Fixed (inflexible) layout: a change of product needs costly retooling
- **Line is only as strong as its weakest station**: one breakdown stops the entire line (no buffer)
- Needs skilled maintenance staff and spare tooling

Key idea: A flow line is a compromise between high production rate and loss of production caused by breakdowns of any one station.

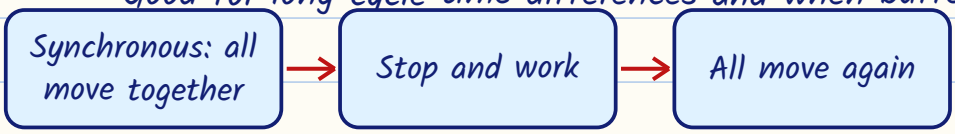
2. Part Transfer Methods and Mechanisms

Two Basic Methods

Part transfer system moves the work-parts from one station to the next. It also **locates and clamps** the part at the station. Two categories of motion exist:

- **Continuous transfer** - work-parts move continuously at constant speed along the line
  - Used in **non-synchronous / bottling / packaging / conveyor assembly** work
  - Workheads move with the part (tracking) or the operator works on the moving part

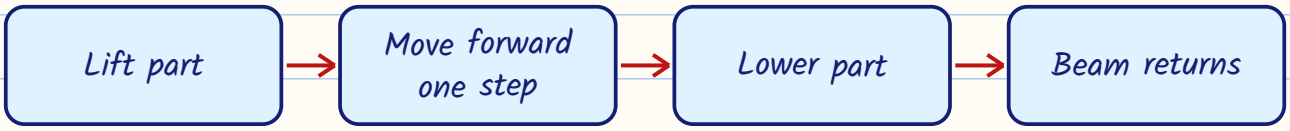
  - Simple, smooth, low shock loads
- **Intermittent (synchronous) transfer** - all parts move together, stop, work is done, then all move again
  - Used in **machining and automatic assembly lines**
  - Part is stationary while being processed, so precise location is possible
  - Motion time = transfer time T_r, then processing time
- **Asynchronous (power-and-free) transfer** - each part moves independently when its station is free
  - Parts can wait in small queues between stations; stations are not locked together
  - Good for long cycle time differences and when buffering is needed



Transfer Mechanisms - Linear Transfer

Used for in-line (straight) lines, parts move along the line.

- **Walking beam** - parts rest on fixed rails; a moving beam lifts, advances and lowers all parts one position. Very accurate, used for heavy parts.
- **Powered roller conveyor** - parts or pallets on driven rollers; simple and cheap.
- **Chain / belt conveyor** - parts move on a belt or chain, either continuously or in steps.
- **Overhead trolley / rail-guided carriers** - hanging parts, painting and plating lines.
- **Power-and-free** system - overhead rail with a power chain and a free trolley that can be released and stopped.
- **Cross-bar transfer** (shuttle bar) - a bar with fingers engages the pallet and pushes it forward; hydraulic or pneumatic drive.



Walking beam: parts are lifted, moved and set down exactly at the stations, so locating accuracy is very good.

Transfer Mechanisms - Rotary Transfer (Indexing)

Parts are carried on a circular table or dial that rotates in steps (indexes).

- **Rack and pinion** - linear reciprocating rack drives a pinion giving rotary motion; simple but limited speed.
- **Ratchet and pawl** - a pawl pushed by a cylinder advances the ratchet wheel by one tooth each stroke.
- **Rotary cam mechanism** - a barrel cam turns the table smoothly; accurate and fast.
- **Geneva mechanism** (Maltese cross) - a driving pin enters a slot in the driven wheel and turns it by one step while the crank turns, then the wheel locks and dwells.

Geneva Mechanism - Analysis

Let n = number of slots in the Geneva wheel. The driven wheel turns one slot angle of 360/n for each full rotation of the driving crank.

$$\text{Angle indexed per cycle} = 360 / n \text{ degrees}$$

$$\text{Crank angle during indexing} = 180 - 360 / n$$

$$T_r = (1/2 - 1/n) T_c$$

- Dwell fraction = 1 - Tr/Tc = 1/2 + 1/n; larger n gives longer dwell. Example: n = 6, Tc = 6 s, so Tr = (0.5 - 0.1667) x 6 = 2 s and 4 s dwell.
- Advantages: smooth start and stop, positive locking during dwell,

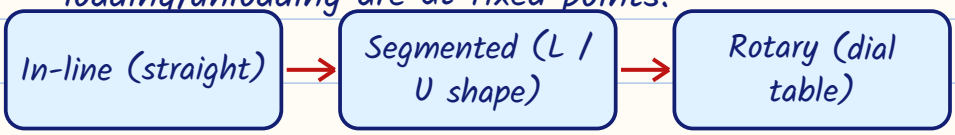
compact, reliable.

- Disadvantages: fixed index angle, jerk at pin entry for low n, limited to light and medium loads.

3. Types of Flow Lines

3.1 Based on Layout (Configuration)

- **In-line** - stations arranged in a straight line; the most common machining transfer line. Easy extension, good access for maintenance.
- **Segmented in-line** - two or more straight segments joined at right angles (L, U, rectangular). Saves floor space; operators can handle several segments.
- **Rotary** - stations around a circular dial table; work-parts are indexed. Compact and rigid, but the number of stations is limited and loading/unloading are at fixed points.



3.2 Based on Storage

- **Without buffer storage** - rigid line; all stations stop when any one stops
- **With buffer storage** - line is divided into **stages** by buffers; a failure in one stage does not stop the others immediately

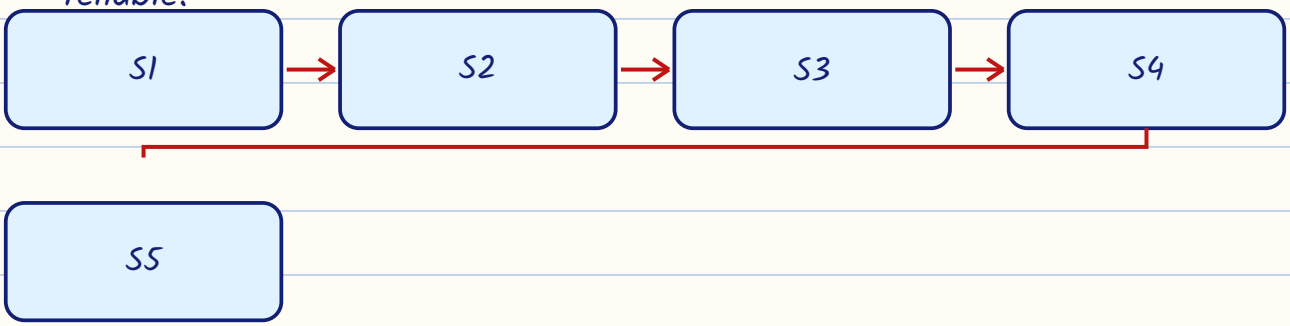
Choose in-line for long, large-volume machining lines; rotary for small parts and fewer operations.

4. Flow Line With and Without Buffer Storage

4.1 Without Buffer (Pure Series Line)

- All stations are linked in strict series; a breakdown at **any one** station stops all other stations.

- The line has the **lowest** **efficiency** but the lowest cost and simplest controls.
- Suitable when the number of stations is small and each station is highly reliable.

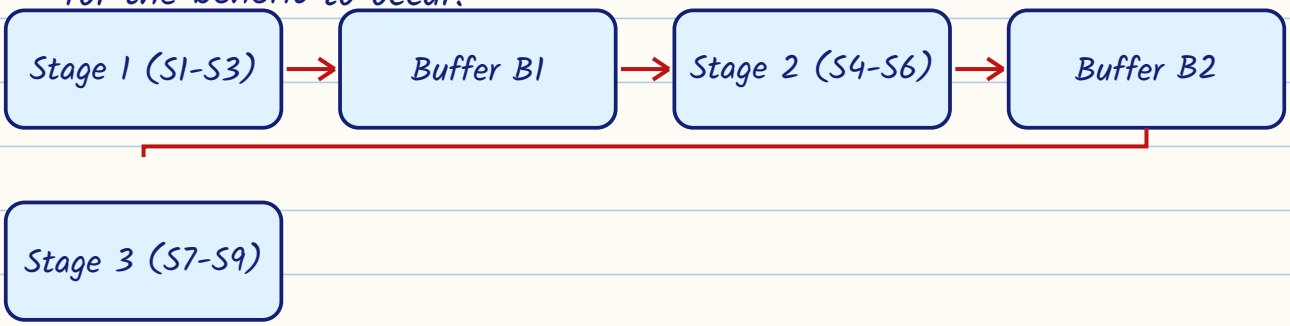


Failure of S3 means S1, S2, S4 and S5 are all forced to stop (no storage to absorb the stop).

4.2 With Buffer Storage

A **buffer** **storage** is a place between stages that stores work-parts temporarily so that the line downstream or upstream can keep running while one stage is down.

- When an upstream stage fails, downstream stages continue to work using parts from the buffer.
- When a downstream stage fails, upstream stages continue and fill up the buffer.
- Buffer must be **neither** **empty** **nor** **full** at the time of the breakdown for the benefit to occur.



Effects and Design Questions

- Larger buffer gives higher efficiency but extra **cost, floor space and WIP**
- Buffer locations: usually between stages of nearly equal reliability; sometimes before expensive or unreliable stations
- Efficiency improvement is **not linear**: first few buffer units give the largest gain; infinite buffer gives the **upper limit**

$$E_0 <= E_b <= E_{infinity}$$

5. Quantitative Analysis of Flow Lines

5.1 Notation

Symbol	Meaning
n	Number of workstations on the line
T_c	Ideal cycle time = max T_{si} + T_r (min)
T_{si}	Service time of station i, T_r = transfer time
F	Breakdown frequency of the line (stoppages per cycle)
T_d	Average downtime per breakdown (min)
T_p	Actual average production cycle time (min)
R_p	Production rate (pieces per min or hour)
E	Line efficiency = proportion of uptime
p_i	Probability that station i breaks down per cycle

5.2 Ideal Cycle Time

The line cycle time is decided by the **slowest station** plus the transfer time.

$$T_c = \max (T_{s1}, T_{s2}, \dots , T_{sn}) + T_r$$

- Ideal production rate (no breakdowns): $R_c = 1 / T_c$

5.3 Breakdowns of the Line

- Every station can stop due to tool failure, jam, adjustment, power failure,

etc.

- F = sum of station breakdown frequencies = $p_1 + p_2 + \dots + p_n$

$$F = \Sigma\, p_i \ (= n\,p \text{ , if all } p \text{ are equal})$$

- Downtime per cycle = $F\,T_d$

5.4 Average Production Cycle Time and Rate

$$T_p = T_c + F\,T_d$$

$$R_p = 1 / T_p = 1 / (T_c + F\,T_d)$$

5.5 Line Efficiency and Downtime Proportion

$$E = T_c / T_p = 1 / (1 + F\,T_d / T_c)$$

$$D = F\,T_d / T_p = 1 - E$$

- E is the fraction of time the line is productive; D is the fraction lost to breakdowns.
- Actual production rate in pieces per hour: $R_p = 60\,E / T_c$

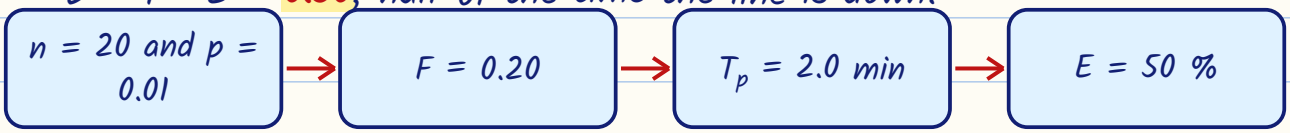
Remember: higher F or higher downtime lowers E. With n stations each with p, $F = n\,p$, so efficiency falls quickly as the number of stations grows.

Worked Example 1 (Series line without buffer)

A 20-station transfer line has $T_c = 1.0$ min and each station has breakdown probability $p = 0.01$ per cycle. Average downtime is $T_d = 5$ min. Find F, T_p, R_p, E and D.

- $F = n\,p = 20 \times 0.01 =$ **0.20 breakdowns/cycle**
- $T_p = T_c + F\,T_d = 1.0 + 0.2 \times 5 =$ **2.0 min**
- $R_p = 1 / 2.0 =$ **0.5 pieces/min = 30/hr**
- $E = T_c / T_p = 1.0 / 2.0 =$ **0.50 = 50 %**

- D = 1 - E = **0.50**; half of the time the line is down.



6. Analysis of Transfer Lines with Buffer Storage

6.1 Upper Bound Approach - Infinite Buffer

Assume a buffer of **infinite capacity** between every pair of stages, so one stage never has to stop because of another. Then each stage's efficiency is its own, and the line efficiency is limited only by the **worst stage**.

$$E_k = 1 / (1 + F_k T_d / T_c) \text{ for stage } k$$

$$E_{infinity} = \min (E_1, E_2, \dots)$$

6.2 Lower Bound - No Buffer

$$E_0 = 1 / (1 + (F_1 + F_2 + \dots) T_d / T_c)$$

6.3 Two-Stage Line with a Finite Buffer

- Stage 1 and stage 2 have frequencies F_1 and F_2, and a buffer of capacity b between them.
- Let $r = F_2 / F_1$; for equal downtime ($T_{d1} = T_{d2}$), the **probability that the buffer is neither empty nor full** is h(b).

$$E_b = E_0 + D_1' \, h(b) \, E_2$$

$$D_1' = F_1 T_d / (T_c + (F_1 + F_2) T_d)$$

$$h(b) = B/(B+1) + (T_c/T_d) \, L / ((B+1)(B+2)) \text{ for } r = 1$$

- Here $b = B (T_d / T_c) + L$, where B is an integer (0, 1, 2 ...) and L is the remainder.

- $h(0) = 0$ (no buffer, no gain) and $h(b)$ approaches 1 as b becomes very large.

Worked Example 4 (Two stages, buffer of 10)

Two equal stages: $F_1 = F_2 = 0.01$, $T_c = 1$ min, $T_d = 5$ min. Buffer capacity $b = 10$.

- $E_0 = 1 / (1 + 0.02 \times 5) =$ **0.909**
- $E_1 = E_2 = 1 / (1 + 0.01 \times 5) =$ **0.952**
- $D_1' = 0.05 / (1 + 0.02 \times 5) = 0.05 / 1.1 =$ **0.0455**
- $b = 10 = B \times 5 + L$ gives $B = 2$, $L = 0$, so $h(b) = 2/3 =$ **0.667**
- $E_b = 0.909 + 0.0455 \times 0.667 \times 0.952 =$ **0.938**

A buffer of only 10 parts recovers about two thirds of the possible gain.

- Place buffers where breakdown frequency is highest; divide the line into stages of roughly equal F; avoid too many buffers (cost, space, WIP).

7. Assembly Line Balancing

7.1 Assembly Process, Systems and Line

Assembly = joining components into a product (40-50 % of production cost). Systems: manual, automated, hybrid, synchronous, asynchronous and flexible assembly lines.

7.2 Assembly Line

An **assembly line** is a series of workstations; the product moves from station to station and workers (or machines) perform a fixed set of tasks at each. Tasks are called **work elements**.

- Work elements have time T_{ek} and **precedence constraints** (what must be done before what)
- Total work content: $T_{wc} = \Sigma\ T_{ek}$

7.3 Terms Used in Line Balancing

- Key terms: T_{wc} total work content, T_c cycle time, T_s station service time, delay time = $T_c - T_s$, precedence diagram, zoning constraint

7.4 Basic Relations

$$T_c = 60\ E / R_p \ (R_p \text{ in units per hour})$$

$$w_{min} = T_{wc} / T_c \ (\text{round up to the next integer})$$

$$E_b = T_{wc} / (w\ T_c) \ (\text{balance efficiency})$$

$$d = 1 - E_b = (w\ T_c - T_{wc}) / (w\ T_c) \ (\text{balance delay})$$

- Balance delay is the **idle time (loss)** because of imperfect balance.
- $E_b = 100\ \%$ only if every station is loaded to the full T_c.

Worked Example 5 (Cycle time and minimum stations)

A product has $T_{wc} = 28$ min. The required output is 40 units/hr and line efficiency is 0.95.

- $T_c = 60 \times 0.95 / 40 =$ **1.425 min**
- $w_{min} = 28 / 1.425 = 19.6$, so **20 workers** are required.

8. Line Balancing Methods

The goal: assign elements to stations to **minimise the number of stations** (or maximise E_b), obeying precedence and cycle-time limits.

8.1 Largest Candidate Rule (LCR)

- List all work elements in descending order of T_e.
- Start with station 1: choose the **largest** element that has no unassigned predecessors and **fits** in the remaining time.
- Repeat until no element fits; open the next station and continue until all elements are assigned.

8.2 Kilbridge and Wester Method (Column method)

- Draw precedence diagram and group elements in **columns** according to precedence (earliest possible column).
- Elements in column I come first, then II, and so on; within the column

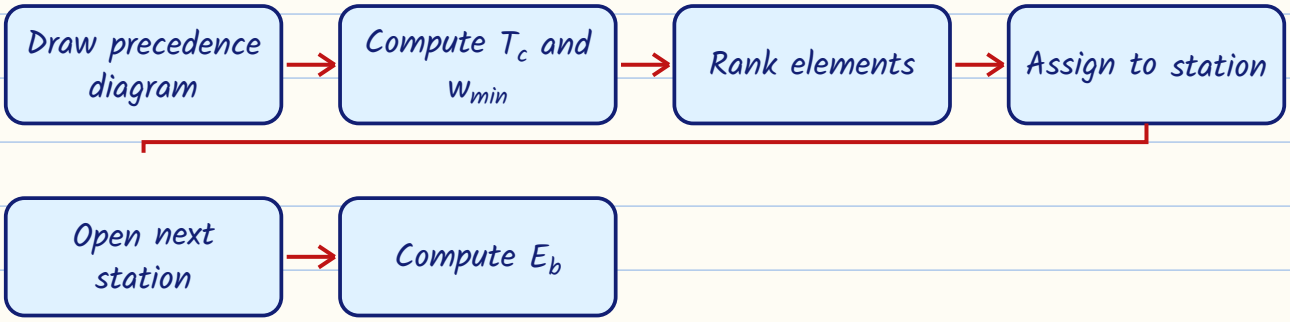
choose by longest time.

- Elements with a choice of column (free) are assigned to fill the gaps.
- Gives a better result than the LCR in many problems because it respects the **position** in the network.

8.3 Ranked Positional Weights (Helgeson and Birnie)

- For each element, **positional weight** RPW = its own time + the times of all elements that **follow** it in the precedence diagram.
- Rank the elements in descending RPW.
- Assign elements to stations in this order if precedence and T_c permit.

$$RPW_i = T_{ei} + \Sigma \text{ (times of all elements after } i \text{)}$$



Worked Example 6 (Largest Candidate Rule)

Elements (time in min, predecessors): 1 (0.5, none), 2 (0.3, after 1), 3 (0.8, after 1), 4 (0.2, after 2), 5 (0.6, after 3), 6 (0.4, after 4 and 5).

Required T_c = 1.0 min.

Element	Time	Predecessor
1	0.5	-
2	0.3	1
3	0.8	1
4	0.2	2
5	0.6	3
6	0.4	4, 5

- T_{wc} = 0.5 + 0.3 + 0.8 + 0.2 + 0.6 + 0.4 = **2.8 min**, w_{min} = 2.8 / 1.0 =

2.8, so **3 stations**

- Station 1: element 1 (0.5); candidates 2 and 3; 3 does not fit (1.3), so take 2 (0.8), then 4 (1.0). **S1 = {1, 2, 4}**, $T_{s1} = 1.0$
- Station 2: element 3 (0.8); element 5 would give 1.4, no fit. **S2 = {3}**, $T_{s2} = 0.8$
- Station 3: elements 5 (0.6) and 6 (0.4). **S3 = {5, 6}**, $T_{s3} = 1.0$

$$E_b = 2.8 / (3 \times 1.0) = 0.933 = 93.3\ \%$$

- Balance delay d = 6.7 %. Idle time exists only at station 2 (0.2 min).
- Same data by RPW: weights 1 = 2.8, 3 = 1.8, 5 = 1.0, 2 = 0.9, 4 = 0.6, 6 = 0.4 gives the same 3-station result (93.3 %).

9. Ways of Improving Line Balance

- **Divide long elements** into smaller ones so that they fill the gaps
- **Reassign** elements between neighbouring stations or **rearrange** precedence where possible
- **Use parallel workstations** for the bottleneck element (two stations doing the same task)
- Use **zoning of similar tasks** / multi-skilled operators who can move to help the busiest station
- Provide **mixed-model sequencing** to level the load across models
- **Reduce the cycle time** by using faster tools, jigs and fixtures, better workplace layout
- **Use underloaded stations** for additional tasks (inspection, packing)
- Use **buffer stock** between stations and **utility workers (floaters)** to cover absence
- Apply **work sampling and time study** to measure real element times, then re-balance periodically

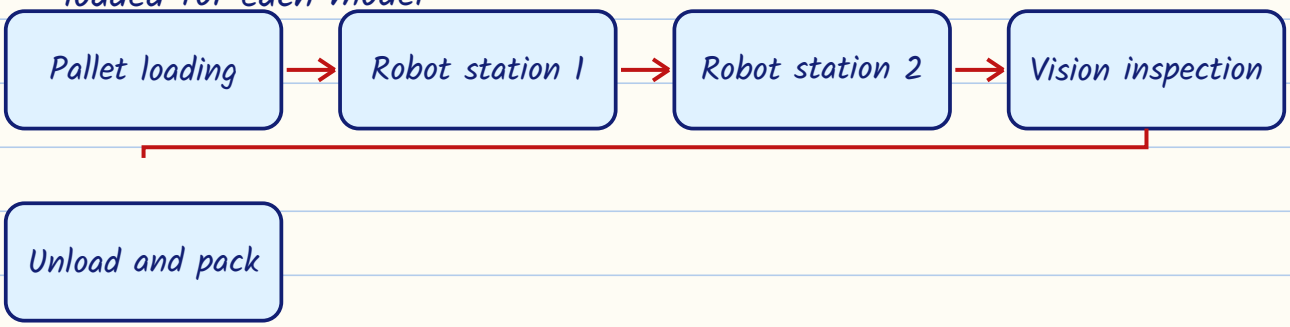
Remember: perfect balance (E_b = 100 %) is rare, so aim at **95 % or above** with minimum workstations.

10. Flexible Assembly Lines

Meaning

A **flexible assembly line (FAL)** consists of programmable assembly stations (robots, NC machines) linked by an automatic transport system, able to assemble **different products or variants** in small or mixed batches without major changeover.

- Stations: assembly robots (SCARA, Cartesian), pick-and-place units, screwdriving and vision inspection
- Transport: pallets on a conveyor, AGVs, shuttle systems with part identification (barcode, RFID)
- Control: central computer with **workstation controllers**; programmes loaded for each model



Features

- Flexibility for **mixed model** production and design changes
- Short setup, quick response to demand
- Higher quality and repeatability; reduced labour
- Integration with **CAD/CAM and ERP**
- Applications: electronic goods (PCBs, relays, motors), automobile parts, domestic appliances

11. Quick Revision Points

- Flow line = stations in series + automatic transfer; **synchronous** (stop-and-go), **asynchronous**, **continuous** transfer
- Mechanisms: walking beam, conveyor, cross-bar (linear); rack-pinion, ratchet-pawl, cam, **Geneva** (rotary)
- Layouts: in-line, segmented in-line, rotary; with or without buffers

$$T_c = \max T_{si} + T_r \mid T_p = T_c + F\,T_d \mid E = T_c / T_p$$

$$R_p = 1 / T_p \mid F = \Sigma\, p_i \mid D = 1 - E$$

- Buffer: $E_0 <= E_b <= E_{infinity}$; $h(b)$ = probability buffer neither empty nor full
- Line balancing: $T_c = 60\,E / R_p$; $w_{min} = T_{wc} / T_c$; $E_b = T_{wc} / (w\,T_c)$; $d = 1 - E_b$
- Methods: **LCR**, Kilbridge-Wester, Ranked positional weights**

Important Exam Questions

- 1. Define an automated flow line and explain its types with neat sketches. (5-10 marks)
- 2. Explain the part transfer mechanisms: walking beam, Geneva, rack and pinion, cam. (10 marks)
- 3. Differentiate a flow line with and without buffer storage; derive E for no buffer. (10 marks)
- 4. A 25-station line has $T_c = 0.8$ min, $p = 0.01$ and $T_d = 4$ min. Find F, T_p, R_p, E. (Answer: $F = 0.25$, $T_p = 1.8$, $R_p = 33.3$/hr, $E = 44.4$ %)
- 5. Explain the upper and lower bound analysis for a two-stage line with a buffer.
- 6. Explain line balancing methods with examples: LCR, Kilbridge-Wester, RPW. (10 marks)
- 7. How can the balance of an assembly line be improved? Write a note on flexible assembly lines.

UNIT - III
Industrial Robotics, Actuators & Sensors

Automation and Robotics (23A03606) - handwritten study notes

1. Introduction to Industrial Robotics

1.0 Syllabus topics covered

- Industrial robotics, configurations, functional line diagram, DOF, arms, joints, grippers, gripper design
- Actuators: pneumatic, hydraulic, electric, stepper; position, velocity, tactile and proximity sensors

1.1 What is a robot?

- The word **robot** comes from the Czech word **robota** meaning forced labour. It was used by Karel Capek in his play R.U.R. (1920)
- The word **robotics** was coined by Isaac Asimov (1940s) who also gave the three laws of robotics
- **RIA definition (Robot Institute of America):** A robot is a reprogrammable, multifunctional manipulator designed to move materials, parts, tools or specialised devices through variable programmed motions for the performance of a variety of tasks.
- **ISO 8373 definition:** An automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which may be fixed in place or mobile, for use in industrial automation applications.

Three key words in every definition: **reprogrammable**, **multifunctional** and **manipulator**. A fixed-sequence machine with no reprogramming is NOT a robot.

1.4 Basic terms

Term	Meaning
Manipulator	Arm and wrist assembly made of links and joints that positions the end effector
End effector	Tool or gripper fixed at the wrist to do the work
Payload	Maximum weight the robot can carry at full speed and still meet its specification
Work envelope	Set of all points in space that the wrist/end effector can reach
Reach	Maximum distance from the base axis to the wrist
Repeatability	How closely the robot returns to a previously taught point (smaller value is better)
Accuracy	How close the robot gets to a commanded point in space
Resolution	Smallest incremental move the control system can command
Speed	Maximum velocity of the tool centre point (TCP)

2. Classification of Robots

3. Classification of Robot Configurations

The first three joints (the **arm** **and** **body**) decide the shape of the work envelope. The last three joints (the **wrist**) decide the orientation of the tool. Joints are either linear (L or P - prismatic) or rotary (R - revolute).

3.1 Cartesian (rectangular / gantry) - LLL or PPP

- Three linear axes X, Y, Z at right angles to each other
- Work envelope is a **rectangular** **box**
- Advantages: simple control, very high accuracy and rigidity, easy to program, large payload in gantry type
- Limitations: needs a lot of floor space, small reach inside the machine, axes are hard to seal
- Uses: pick and place, CNC loading, assembly, 3D printing

3.3 Spherical (polar) - RRL (TRL)

- One linear telescoping arm that can pivot about a horizontal axis and rotate about a vertical base axis
- Work envelope is a **part of a sphere**
- Advantages: long reach, can reach down to the floor level
- Limitations: low accuracy at long extension, complex control
- Uses: die casting, forging (Unimate and Versatran were polar robots)

3.4 Articulated (jointed arm / anthropomorphic) - RRR

- Three rotary joints: waist, shoulder and elbow, like a human arm
- Work envelope is an **irregular sphere-like volume**
- Advantages: maximum flexibility, compact, large work volume, can reach over and around obstacles
- Limitations: complex control and kinematics, low stiffness at full stretch, accuracy changes with position
- Uses: arc welding, spray painting, assembly - the most common industrial robot

3.7 Comparison of configurations

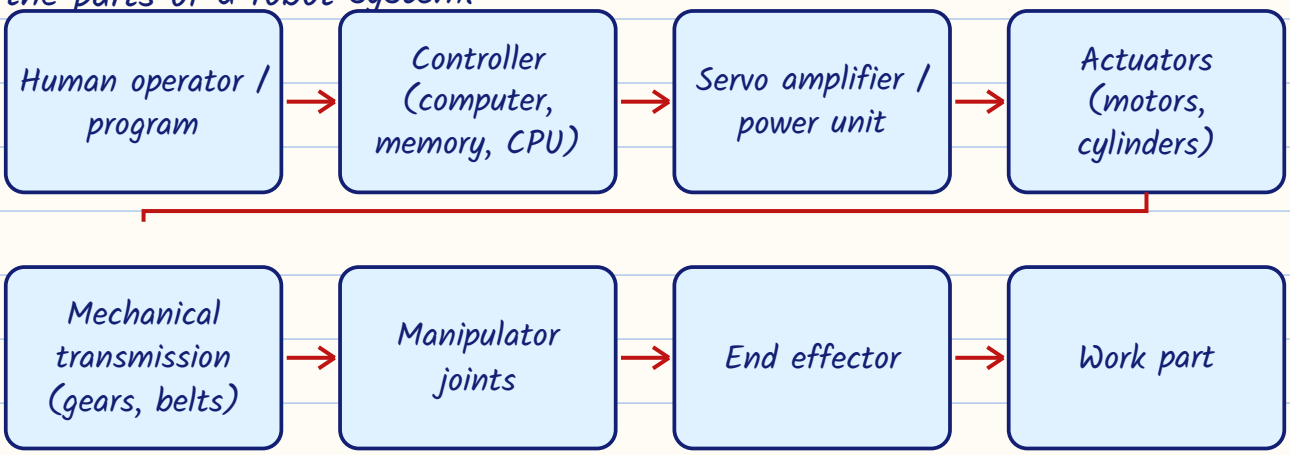
Type	Joints	Work envelope	Main use
Cartesian	PPP	Rectangular box	Pick and place, gantry
Cylindrical	RPP	Cylinder	Machine loading
Spherical	RRP	Sphere segment	Forging, die casting
Articulated	RRR	Irregular sphere	Welding, painting
SCARA	RRP	Cylinder-like (horizontal)	Assembly

Cartesian X Y Z → Cylindrical R Z r → Spherical R R r → Articulated R R R

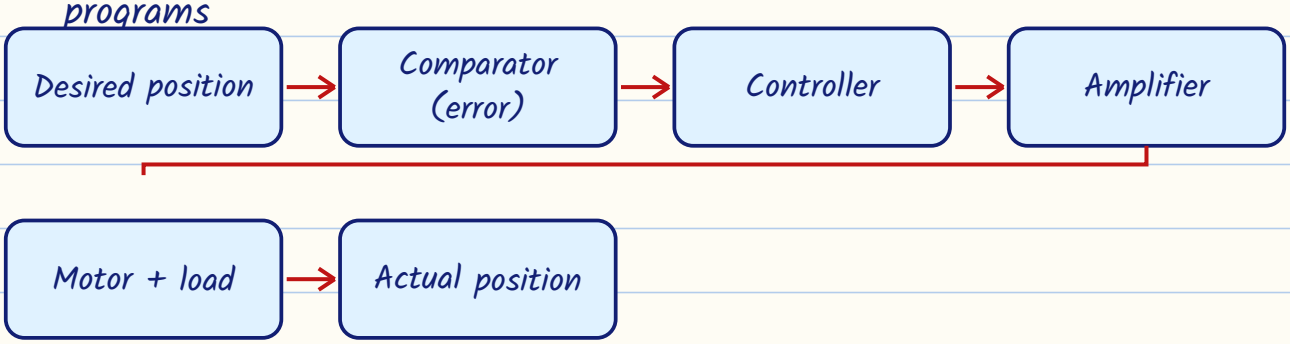
Memory trick: the number of rotary joints rises from 0 to 3 as we go Cartesian (0) ⟶ Cylindrical (1) ⟶ Spherical (2) ⟶ Articulated (3).

4. Functional Line Diagram of a Robot

The functional line diagram shows how information and energy flow between the parts of a robot system.



- Feedback path: **sensors** on every joint measure position and speed and return the signal to the controller. This closed loop gives accuracy
- Position sensors (encoder, resolver, potentiometer) give joint angle; tachogenerator gives velocity
- External sensors (vision, force, touch, proximity) tell the controller about the work environment
- The teach pendant is used by the operator to teach points and edit programs



- The comparator subtracts the **feedback** signal from the **command** signal. The error drives the motor until the error is zero

> A robot with feedback is a closed loop (servo) system. A robot without feedback (stepper driven, pneumatic stops) is an open loop system.

5. Degrees of Freedom (DOF)

Degree of freedom is an independent motion (translation or rotation) that a mechanism can make. Each joint of a robot gives it one DOF (one axis). In space a free rigid body has **6 DOF**: three translations (x, y, z) and three rotations (roll, pitch, yaw).

- To place a tool at any position and orientation in space a robot needs at least **6 DOF**
- First 3 DOF (arm and body) position the wrist. Last 3 DOF (wrist) orient the tool
- A robot with more than 6 DOF is called **redundant**, it can avoid obstacles. A robot with fewer DOF can reach only some orientations

6. Components - Common Types of Arms and Joints

6.5 The wrist

- Wrist is made of 2 or 3 rotary joints with intersecting axes: **roll, pitch, yaw** (RPY)
- Types: 2 DOF wrist (pitch and roll), 3 DOF wrist (roll-pitch-roll or roll-pitch-yaw)
- The wrist must be light because it sits at the end of the arm and its weight is multiplied by the arm length
- A **spherical wrist** (three axes meet at one point) makes inverse kinematics simple

7. Grippers (End Effectors)

End effector is the device attached to the wrist of the robot to perform the task. It is of two types: **grippers** (hold objects) and **tools** (weld torch, spray gun, drill, grinder).

7.1 Types of grippers by actuation / principle

- **Mechanical grippers** - fingers move to grasp the part (parallel or angular motion)
- **Vacuum (suction) grippers** - suction cups hold flat smooth parts such as glass, sheet metal and boxes
- **Magnetic grippers** - electromagnet or permanent magnet holds ferrous parts
- **Adhesive grippers** - sticky surface for light fabric type parts
- **Hooks, scoops and ladles** - for lifting and for liquids or powders
- **Expandable (inflatable) grippers** - bladder expands in a hole or around a part

7.3 Gripping force

For a friction grip with coefficient of friction mu and number of contact fingers n:

$$F = (m\ g\ a) / (mu\ n)$$

- m = mass of part, g = 9.81, a = acceleration factor (safety factor, usually 2 to 3)
- The force must be large enough to stop the part slipping but small enough not to damage it

7.4 Worked example

Q. A part of mass 5 kg is held by two fingers with friction coefficient 0.25. The part is lifted vertically with an acceleration factor g_factor = 2. Find the gripping force needed per finger.

- Weight W = m g = 5 x 9.81 = 49.05 N
- Using F = W x factor / (mu x n) = 49.05 x 2 / (0.25 x 2) = **196.2 N**

per finger

8. Factors in the Design of Grippers

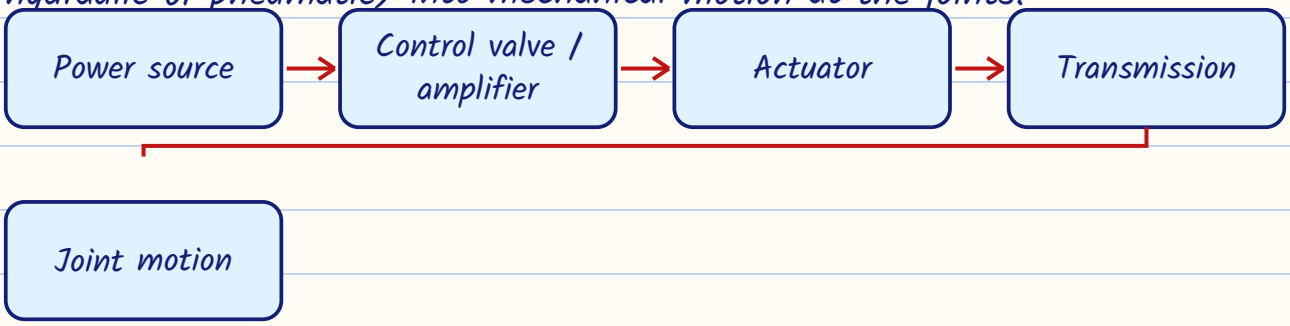
The gripper must hold the part firmly and positively during the entire motion without damaging it, and release it quickly at the right place.

- **Weight and size of the part** - decide the force and the gripper size
- **Shape of the part** - round, flat, prismatic; fingers are shaped to match (V-shaped fingers for cylinders)
- **Gripping force** - enough to overcome weight, inertia and friction with a safety factor
- **Part surface** - hot, wet, oily, soft or fragile parts need special pads
- **Gripper weight** - should be low, the payload of the robot includes the gripper weight
- **Approach and clearance** - fingers must enter and leave without hitting other parts
- **Positive grip** - the part should be located by shape (self centring) so that the position is known
- **Change of part size** - interchangeable fingers or adjustable stroke for part variety
- **Fail safe** - the part must not drop if power fails (spring close, vacuum reservoir)
- **Environment** - temperature, dust, chemicals and explosive atmosphere
- **Cost, reliability and ease of maintenance**
- **Speed of operation** - open and close time must fit within the cycle
- **Use of sensors** - touch or proximity sensors in the fingers to detect part presence and force

9. Robot Actuators and Feedback Components

9.1 Actuators

Actuators are the muscles of the robot. They convert energy (electrical, hydraulic or pneumatic) into mechanical motion at the joints.



- Requirements: high power to weight ratio, high torque at low speed, quick response, smooth motion, accurate control, good efficiency, low cost

9.2 Pneumatic actuators

- Use compressed air at 5 to 8 bar, supplied by a compressor
- Parts: air cylinder (single or double acting), rotary vane motor, directional control valves, FRL unit (filter, regulator, lubricator)
- Mostly used with a bang-bang (end stop) control, so used for pick and place robots and gripper opening
- Advantages: clean, cheap, fast, simple, safe in explosive areas, air is available everywhere
- Limitations: air is compressible so position control is poor, low force, noisy exhaust, needs air preparation

9.3 Hydraulic actuators

- Use mineral oil at 70 to 200 bar from a pump, reservoir and servo valve
- Types: linear cylinders, rotary vane actuators, hydraulic motors (gear, vane, piston)
- Advantages: very high force and torque, high power to weight ratio, stiff, good speed control, self lubricating, can hold position under load
- Limitations: leakage and mess, needs pump, filter and cooler, noise, fire hazard, high maintenance, expensive
- Used in heavy robots such as forging and big painting robots

9.4 Electric actuators

- **DC servo motors** - permanent magnet DC motor, speed controlled by armature voltage, torque is proportional to armature current
- **Brushless DC (BLDC) motors** - electronic commutation, no brush wear, high reliability, widely used now
- **AC servo motors** - induction or synchronous type, driven by variable frequency drives
- **Stepper motors** - move in fixed angular steps for each pulse (see next section)
- Advantages: clean, quiet, accurate, easy to control by computer, good efficiency, low maintenance
- Limitations: lower power to weight than hydraulic, needs gear reducers, sparking in brush types

10. Stepper Motors

A **stepper motor** is a brushless motor that divides a full rotation into a number of equal steps. Each pulse from the driver rotates the shaft by exactly one **step angle**. It runs as an open loop with no feedback sensor.

Step angle = 360 / (number of steps per revolution)

10.1 Types

- **Variable reluctance (VR)** - soft iron toothed rotor, no permanent magnet, step angle 15 degrees or so, no holding torque when off
- **Permanent magnet (PM)** - magnet rotor, bigger steps 45 or 90 degrees, has detent torque
- **Hybrid** - combines both, step angle 1.8 or 0.9 degree, the most common type in robots and CNC

10.3 Speed and position

N (rpm) = (step angle x pulse frequency x 60) / 360

- Position (angle) = number of pulses x step angle
- **Example:** step angle 1.8 degrees, pulse frequency 1000 pulses per second. Steps per revolution = 360 / 1.8 = 200. N = 1000 / 200 x 60 = **300 rpm**
- **Example:** To rotate the shaft by 90 degrees we need 90 / 1.8 = **50 pulses**

10.4 Advantages and limitations

Advantages	Limitations
Open loop control, no sensor needed	Can lose steps if overloaded - error is not known
Accurate positioning and good repeatability	Low torque at high speed, torque falls with speed
Holding torque at standstill	Resonance and vibration at some speeds
Cheap and simple digital control	Low power - used for small robots only
Long life, no brushes	Draws full current even when idle, heating

11. Comparison of Actuators

Feature	Pneumatic	Hydraulic	Electric
Power source	Air 5 to 8 bar	Oil 70 to 200 bar	Electric supply
Power to weight	Low	Very high	Medium
Payload	Small	Very large	Small to medium
Accuracy	Poor	Good	Very good
Speed	High	Medium	High
Stiffness	Low (air compressible)	High	Medium (depends on gearing)
Control	Simple stops	Servo valve	Easy, digital
Cleanliness	Clean	Oil leakage	Very clean
Noise	Exhaust noise	Pump noise	Quiet
Maintenance	Low	High	Low

Cost	Low	High	Medium
Typical use	Pick and place, grippers	Heavy handling, forging	Welding, assembly, general use

Selection rule: light and fast pick and place - pneumatic; very heavy loads - hydraulic; precision and general work - electric servo.

12. Feedback Components - Sensors in Robotics

Sensors give the controller information. **Internal sensors** measure the state of the robot itself (position, velocity, acceleration). **External sensors** measure the environment (touch, proximity, vision, force).

13. Position Sensors

13.1 Potentiometer

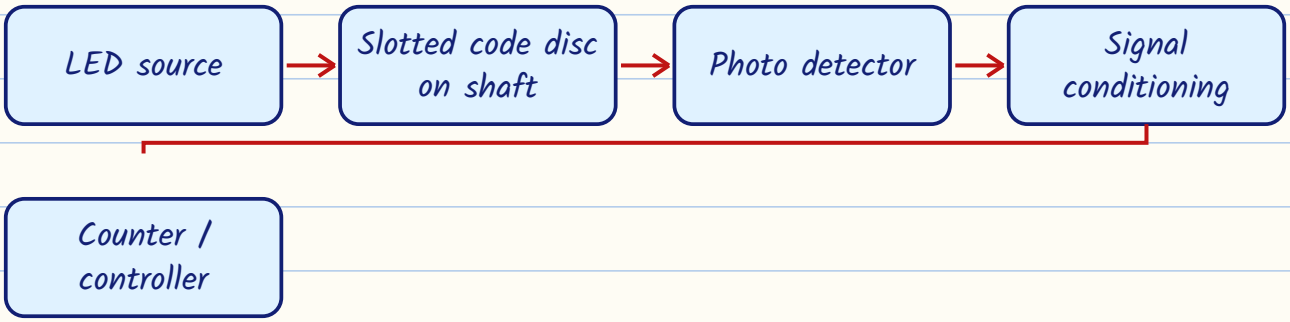
- A resistive track with a sliding contact (wiper) attached to the joint. Output voltage is proportional to the position

$$V_{out} = V_s \times (x / L)$$

- Rotary type for angular and linear type for sliding joints
- Advantages: simple, cheap, analog output directly proportional to position
- Limitations: wear of the wiper and track, noise, limited life, low resolution, friction adds to torque
- **Example:** supply 10 V, a rotary pot with 300 degree travel is at 90 degrees. Vout = 10 x 90 / 300 = **3 V**

13.2 Optical encoders

An encoder is a digital transducer that converts shaft rotation into electrical pulses. It has a **disc with slots**, an **LED light source** and a **photo detector**.



Incremental encoder

- Disc has equally spaced slots. Each slot passes gives one pulse. Counting pulses gives the angle
- Two tracks A and B are shifted by 90 degrees (**quadrature**). The phase tells the direction. A third track Z gives one index pulse per revolution (home mark)
- Resolution = 360 / N degrees per pulse (N = slots). With quadrature counting the resolution is 4 times better
- Limitation: loses the position when power is off - needs a homing routine at start

Absolute encoder

- Disc has multiple concentric tracks coded in **binary or Gray code**. Each position has a unique code
- Resolution = 360 / 2^n for n tracks. Example: 10 tracks = 1024 positions = 0.35 degree
- Advantage: position is known immediately at power on. Limitation: costly and complex
- **Gray code** changes only one bit at a time so reading errors are avoided

Example: A 1000 slot incremental encoder with 4x quadrature gives 4000 counts per rev, so the resolution is 360 / 4000 = 0.09 degree.

13.3 Resolver

- A resolver is an electromagnetic rotary transformer, like a small AC machine, with one rotor winding and two stator windings placed 90

degrees apart

- The rotor is fed with an AC reference voltage V sin(wt). The stator outputs are:

$$V_1 = V \sin(wt) \cos(theta),\ V_2 = V \sin(wt) \sin(theta)$$

- The angle is found from the ratio: theta = $\tan^{-1}(V_2 / V_1)$, done by a resolver to digital converter
- Advantages: very rugged, works at high temperature, vibration, dust and oil. Absolute within one turn, long life, no contact wear
- Limitations: needs AC excitation and an RDC circuit, lower resolution than a good encoder, costly

14. Velocity Sensors

- Needed in the servo loop for speed feedback and to damp the motion (the derivative term in PID)

14.1 DC tachogenerator

- A small DC (or AC) generator coupled to the motor shaft. Output voltage is proportional to speed

$$V = K_g \times N$$

- Typical constant 3 to 7 V per 1000 rpm. Simple analog output but brush wear and ripple

15. Tactile Sensors

Tactile (touch) sensors respond to physical contact, force or pressure between the gripper and the object. They give the robot a sense of touch.

15.1 Types

- **Touch sensors** - on/off only (binary). Limit switch, micro switch, whisker that detects contact
- **Force sensors** - measure the size of the force or pressure

- **Slip sensors** - detect the slip of a held object so the grip can be increased

15.2 Principles

- **Strain gauge** - the resistance of a thin wire or foil changes with strain. Bonded to a wrist or finger element and read in a Wheatstone bridge
- **Piezoelectric** - crystals like quartz generate a charge under force. Good for dynamic forces
- **Piezoresistive / conductive rubber** - the resistance of the rubber falls when pressed. Cheap and flexible; used in artificial skin
- **Capacitive** - pressing changes the gap between plates and the capacitance
- **Optical** - pressed rubber changes the light in a fibre or reflection

16. Proximity and Range Sensors

Proximity sensors detect the presence of a nearby object without physical contact. **Range sensors** also measure the distance.

16.1 Inductive proximity sensor

- A coil produces a high frequency magnetic field. A metal object near it causes eddy currents and changes the oscillation, which switches the output
- Detects **metals only**, range 1 to 30 mm, robust against dirt and oil

16.4 Ultrasonic sensor

- Emits a sound pulse of 40 kHz and measures the time of the echo. Distance is found from time of flight

$$d = (v \times t) / 2$$

- Speed of sound v = 343 m/s. **Example:** an echo returns after 5.8 ms. d = 343 x 0.0058 / 2 = **0.995 m, about 1 m**
- Works on any material and colour but has a wide beam and a blind zone near the sensor

17. Quick Revision

- Robot = reprogrammable multifunctional manipulator. 6 DOF are needed for an arbitrary position and orientation
- Configurations: Cartesian PPP, cylindrical RPP, spherical RRP, articulated RRR, SCARA RRP
- Gripper types: mechanical, vacuum, magnetic, adhesive; the design depends on the part weight, shape, force and safety
- Actuators: pneumatic (fast, cheap), hydraulic (powerful), electric (accurate, clean). Stepper = open loop, step angle = 360 / steps
- Position: potentiometer, incremental encoder, absolute encoder, resolver, LVDT. Velocity: tachogenerator, encoder
- Tactile: strain gauge, piezo, conductive rubber. Proximity: inductive, capacitive, optical, ultrasonic

17.1 Important questions

- Define a robot. Explain the main components of an industrial robot with a neat sketch
- Classify robots by configuration. Compare with work envelopes
- What is meant by degrees of freedom? Explain the six DOF of a robot
- Explain the types of joints and the common types of robot arms
- Explain different types of grippers. What factors must be considered while designing a gripper?
- Compare pneumatic, hydraulic and electric actuators
- Explain the working of a stepper motor. Why is it used in open loop control?
- Explain the working of incremental and absolute encoders. Compare them with the resolver
- Explain tactile sensors and proximity sensors used in robots
- A stepper motor has 200 steps per revolution and is driven at 500 pulses per second. Find the speed in rpm (Ans 150 rpm)

UNIT - IV

Manipulator Kinematics, Dynamics & Trajectory Planning

Automation and Robotics (23A03606) - handwritten study notes

Syllabus Topics

- Manipulator Kinematics: homogeneous transformations (rotation and translation), D-H notation, forward and inverse kinematics
- Manipulator Dynamics: differential transformations, Jacobians, Lagrange-Euler and Newton-Euler formulations
- Trajectory Planning: obstacle avoidance, path planning, skew motion, joint integrated motion, straight line motion

Why this unit matters: kinematics tells us WHERE the hand is for given joint angles (and vice versa), dynamics tells us WHAT forces/torques are needed, trajectory planning tells us HOW to move smoothly between points.

1. Manipulator Kinematics

1.1 What is kinematics?

Kinematics is the study of motion of the manipulator **without considering the forces** that cause the motion. It deals with position, velocity and acceleration of links and the end-effector.

- A manipulator is a chain of **rigid links** connected by **joints** (revolute R or prismatic P).
- Base of the arm is fixed, free end carries the **end-effector** (gripper / tool).
- Two basic problems are solved in kinematics:
  - **Forward kinematics**: joint variables ⟶ position and orientation of the end-effector

– **Inverse kinematics**: desired position and orientation of end-effector ⟶ joint variables

1.2 Forward vs Inverse kinematics

Point	Forward kinematics	Inverse kinematics
Given	Joint angles / displacements	Required end-effector pose
Find	End-effector position and orientation	Joint angles / displacements
Solution	Always unique, straightforward	May have many solutions or none
Method	Matrix multiplication of link transforms	Algebraic, geometric or numerical
Use	Display, simulation, checking	Control, motion programming

Key point: forward kinematics has a **unique** solution; inverse kinematics may have **multiple** solutions (elbow-up / elbow-down) or **no** solution if the point is outside the workspace.

2. Homogeneous Transformations

2.1 Why homogeneous matrices?

Rotation is a 3 x 3 matrix multiplication, but translation is an addition. To combine both into **one single matrix multiplication** we use a 4 x 4 **homogeneous transformation matrix**.

- A point (x, y, z) is written as a 4 x 1 column vector (x, y, z, 1)
- A 4 x 4 matrix T has the form: rotation 3 x 3 (top-left), position vector 3 x 1 (top-right), perspective 1 x 3 = 0 and scale factor = 1 (bottom row)

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} & p_x \\ R_{21} & R_{22} & R_{23} & p_y \\ R_{31} & R_{32} & R_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Advantage**: a chain of transformations is simply the product $T = T_1 T_2$

T_3 ...

- **Caution**: matrix multiplication is **not commutative**, so the order matters.

2.2 Pure translation

Translating a point by (a, b, c) along x, y, z:

$$\begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Written as Trans(a, b, c). Inverse = Trans(-a, -b, -c).

2.3 Pure rotation about the axes

Using the shorthand c = cos(theta), s = sin(theta):

Rotation about X-axis, Rot(x, theta)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c & -s & 0 \\ 0 & s & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about Y-axis, Rot(y, theta)

$$\begin{bmatrix} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about Z-axis, Rot(z, theta)

$$\begin{bmatrix} c & -s & 0 & 0 \\ s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Properties of rotation matrices: columns are **orthogonal unit vectors**, determinant = +1, and the **inverse equals the transpose**: $R^{-1} = R^T$.

2.4 Inverse of a homogeneous matrix

For T with rotation R and position p:

$$T^{-1} = [\ R^T \ -R^T p\ ;\ 0\ 1\ ]$$

- Do not invert a 4 x 4 numerically, use this shortcut. It saves computation time in real-time controllers.

2.5 Composite transformations

- **Rotations about the current (moving) axes**: post-multiply (new transform on the RIGHT).
- **Rotations about the fixed (reference) axes**: pre-multiply (new transform on the LEFT).

Worked example 1

Point P = (7, 3, 2) is rotated 90 degrees about the Z-axis and then translated by (4, -3, 7). Find the new coordinates.

- Step 1: Rot(z, 90): c = 0, s = 1, so $x' = c\,x - s\,y = -3$, $y' = s\,x + c\,y = 7$, $z' = 2$.
- Step 2: Translation: $x'' = -3 + 4 = 1$, $y'' = 7 - 3 = 4$, $z'' = 2 + 7 = 9$.

Answer: P' = (1, 4, 9). (Order: T = Trans(4,-3,7) . Rot(z,90) applied to P.)

3. Degrees of Freedom and Joint Notation

- **Degrees of freedom (DOF)**: number of independent parameters needed to fix the pose of the end-effector. A free body in space has 6 DOF.
- A manipulator needs **6 joints** to reach any position AND orientation in its workspace (3 for arm and body, 3 for wrist).
- Joint types:

- **Revolute (R)**: rotation about an axis, variable = theta
- **Prismatic (P)**: sliding along an axis, variable = d
- Both have 1 DOF each.

4. Denavit-Hartenberg (D-H) Notation

4.1 Idea

Denavit and Hartenberg (1955) proposed a **systematic method** of attaching a coordinate frame to every link so that the relation between two successive links is described by just **four parameters**.

- Joint i connects link i-1 and link i.
- Frame {i} is attached to link i; Z-axis of frame {i-1} lies along the axis of joint i.

4.2 The four D-H parameters

Parameter	Name	Definition
$theta_i$	Joint angle	Rotation about z_{i-1} from x_{i-1} to x_i
d_i	Link offset	Distance along z_{i-1} from x_{i-1} to x_i
a_i	Link length	Distance along x_i from z_{i-1} to z_i
$alpha_i$	Link twist	Angle about x_i from z_{i-1} to z_i

For a **revolute** joint $theta_i$ is the **variable** (other three constant). For a **prismatic** joint d_i is the **variable**.

4.3 Procedure to assign frames (step by step)

- Step 3: Assign z_{i-1} along the axis of joint i.
- Step 5: Assign x_i along the common normal between z_{i-1} and z_i ($x_i = z_{i-1} \times z_i$).
- Step 7: Fill the **D-H table** with theta, d, a, alpha for each link.
- Step 8: Write the **A-matrix** for each link and multiply.

4.4 Link transformation matrix (A-matrix)

Going from frame {i-1} to frame {i} = Rot(z, theta) . Trans(0,0,d) .

Trans(a,0,0) . Rot(x, alpha):

$$\begin{bmatrix} c_i & -s_i c a_i & s_i s a_i & a_i c_i \\ s_i & c_i c a_i & -c_i s a_i & a_i s_i \\ 0 & s a_i & c a_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Here $c_i = \cos(\text{theta}_i)$, $s_i = \sin(\text{theta}_i)$, $ca_i = \cos(\text{alpha}_i)$, $sa_i = \sin(\text{alpha}_i)$.
- Forward kinematics of an n-joint arm:

$$T_0^n = A_1 A_2 A_3 \dots A_n$$

Memorise this matrix. It is asked in almost every exam. For a **6-DOF** arm the result $T = A_1A_2A_3A_4A_5A_6$ gives the hand pose (n, o, a, p).

5. Forward Kinematics

Link lengths L_1 and L_2; joint angles theta_1 and theta_2. Both joint axes are parallel and perpendicular to the plane.

Link	theta	d	a	alpha
1	theta_1	0	L_1	0
2	theta_2	0	L_2	0

Since alpha = 0 and d = 0 each A-matrix becomes:

Multiplying A_1 A_2 and using the identities $\cos(a+b) = \cos a \cos b - \sin a \sin b$, $\sin(a+b) = \sin a \cos b + \cos a \sin b$:

$$\begin{bmatrix} c_{12} & -s_{12} & 0 & L_1c_1+L_2c_{12} \\ s_{12} & c_{12} & 0 & L_1s_1+L_2s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- where $c_{12} = \cos(\theta_1 + \theta_2)$ and $s_{12} = \sin(\theta_1 + \theta_2)$.

End-effector position: $x = L_1\cos\theta_1 + L_2\cos(\theta_1+\theta_2)$ and $y = L_1\sin\theta_1 + L_2\sin(\theta_1+\theta_2)$. Orientation angle $\phi = \theta_1 + \theta_2$.

Numerical example

$L_1 = 2$ m, $L_2 = 1$ m, $\theta_1 = 30$ degrees, $\theta_2 = 60$ degrees.

- $x = 2\cos30 + 1\cos90 = 2(0.866) + 0 =$ **1.732 m**
- $y = 2\sin30 + 1\sin90 = 2(0.5) + 1 =$ **2.0 m**

6. Inverse Kinematics

6.1 Problem and difficulties

Given the desired hand pose (position + orientation) find $\theta_1 \dots \theta_n$. The equations are **non-linear** and **coupled**.

- Problems: **multiple solutions**, **no solution** (out of reach), **singularities**, and no general closed-form for all arms.

6.2 Geometric solution of 2-link planar arm

Target point (x, y). Use the cosine rule on the triangle formed by L_1, L_2 and the line from base to target.

$$x^2 + y^2 = L_1^2 + L_2^2 + 2\,L_1\,L_2\cos(\theta_2)$$

$$\cos(\theta_2) = (\,x^2 + y^2 - L_1^2 - L_2^2\,) / (\,2\,L_1\,L_2\,)$$

- Step 1: compute $\cos(\theta_2)$ as above. If its magnitude is greater than 1, the point is **unreachable**.
- Step 2: $\theta_2 = \arccos(.)$ gives **two solutions** (+ and -): **elbow-down** (+) and **elbow-up** (-).
- Step 3: find $\theta_1 = \text{atan2}(y, x) - \text{atan2}(L_2\sin(\theta_2), L_1 + L_2\cos(\theta_2))$.

> Always use **atan2(y, x)** rather than atan(y/x): it keeps the correct quadrant and avoids division by zero.

Worked example (inverse)

$L_1 = L_2 = 1$ m, target $(x, y) = (1, 1)$.

- $\cos(\theta_2) = (1 + 1 - 1 - 1) / (2) = 0$ so **θ_2 = +90 degrees** (or -90).
- $\theta_1 = \text{atan2}(1, 1) - \text{atan2}(1 \sin 90, 1 + 0) = 45 - 45 =$ **0 degrees**.
- Second solution: $\theta_2 = -90$, $\theta_1 = 45 - \text{atan2}(-1, 1) = 45 + 45 = 90$ degrees.

7. Manipulator Dynamics: Differential Transformations

- A **differential transformation** describes an infinitesimally small translation $d = (dx, dy, dz)$ and rotation $\delta = (\delta_x, \delta_y, \delta_z)$; for small angles $\sin(\delta) = \delta$, $\cos(\delta) = 1$.

> Small rotations about different axes **commute** (order does not matter), unlike finite rotations. Change of a frame: **dT = Delta . T**.

7.3 Differential motion of a manipulator

The change in hand position (dx, dy, dz) and rotation $(\delta_x, \delta_y, \delta_z)$ is related to the change in joint variables dq_i:

$$D = J \cdot dq$$

- $D = [dx\ dy\ dz\ \delta_x\ \delta_y\ \delta_z]^T$ is the **hand differential vector** (6 x 1).
- $dq = [dq_1 \dots dq_n]^T$ is the joint differential vector, J is the **Jacobian**.

8. The Jacobian

8.1 Definition

The **Jacobian** is the matrix that maps **joint velocities** to **end-effector**

(Cartesian) velocities:

$$V = J(q) \cdot q\text{-dot}$$

- $V = [vx\ vy\ vz\ omega_x\ omega_y\ omega_z]^T$ - linear and angular velocity of the hand.
- Size is 6 x n for an n-joint arm. It is **configuration dependent** (changes with q).
- Each column of J = effect of one joint's motion on the hand.

8.2 Jacobian of the 2-link planar arm

Start from the position equations and differentiate:

$$x = L_1c_1 + L_2c_{12}\quad y = L_1s_1 + L_2s_{12}$$

$$\begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix}$$

- so that $[vx\ ;\ vy] = J \cdot [\theta_1\text{-dot}\ ;\ \theta_2\text{-dot}]$

Numerical example

$L_1 = L_2 = 1$ m, $\theta_1 = 0$, $\theta_2 = 90$ degrees. Joint speeds 1 rad/s each.

- $J = [\ -(0) - 1\ ,\ -1\ ;\ 1 + 0\ ,\ 0\ ] = [\ -1\ -1\ ;\ 1\ 0\ ]$
- $vx = (-1)(1) + (-1)(1) =$ **-2 m/s**, $vy = (1)(1) + 0 =$ **1 m/s**

8.3 Uses of the Jacobian

- **Velocity relation** between joint space and Cartesian space.
- **Force relation** by the principle of virtual work: **$\tau = J^T F$** (joint torques from the force at the hand).

8.4 Singularity from the Jacobian

For the planar arm: $\det(J) = L_1 L_2 \sin(\theta_2)$.

- $\det(J) = 0$ when **$\theta_2 = 0$ or 180 degrees** (arm fully stretched or folded back).
- At these points the hand cannot move along the arm direction; the Jacobian **loses rank**.

Key point: near a singularity the inverse Jacobian becomes huge, so joint velocities required are very large. Avoid singular configurations in path planning.

9. Dynamics of Manipulators

9.1 What is dynamics?

Dynamics studies the **relation between the forces/torques applied at the joints and the resulting motion** (positions, velocities, accelerations).

- **Forward dynamics**: given torques find the motion (used in simulation).
- **Inverse dynamics**: given the desired motion find the joint torques (used in control).

General equation of motion

$$\tau = M(q)\,\ddot{q} + C(q, \dot{q})\,\dot{q} + G(q)$$

- τ = joint torque vector, $M(q)$ = **inertia matrix** (symmetric, positive definite), C = **Coriolis and centrifugal** terms, G = **gravity** loading terms.

9.2 Lagrange-Euler (L-E) formulation

An **energy based** method. It uses Lagrangian $L = K - P$ where K = total **kinetic energy** and P = total **potential energy** of the arm.

$$\tau_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i}$$

Steps of L-E method

- Step 1: Assign D-H coordinate frames to all links.
- Step 2: Find the position of the centre of mass of every link, using the A-matrices.
- Step 4: Calculate the **kinetic energy** K of each link: translational + rotational, then sum them.
- Step 7: Differentiate with respect to q_i and $\dot{q}_i$, then with time, to obtain τ_i.

9.3 L-E for a 2-link planar arm

Point masses m_1, m_2 at the ends of links of length L_1, L_2.

- Kinetic energy: $K = (1/2)\ m_1\ v_1^2 + (1/2)\ m_2\ v_2^2$
- Potential energy: $P = m_1\ g\ L_1\ s_1 + m_2\ g\ (\ L_1 s_1 + L_2 s_{12}\ )$

Resulting torque equations (standard form):

$$tau_1 = D_{11}\ theta_1\text{-ddot} + D_{12}\ theta_2\text{-ddot} + (Coriolis\ terms) + D_1$$

$$tau_2 = D_{21}\ theta_1\text{-ddot} + D_{22}\ theta_2\text{-ddot} + (Coriolis\ terms) + D_2$$

- $D_{11} = (m_1+m_2)\ L_1^2 + m_2\ L_2^2 + 2\ m_2\ L_1\ L_2\ c_2$
- $D_{12} = D_{21} = m_2\ L_2^2 + m_2\ L_1\ L_2\ c_2$, $D_{22} = m_2\ L_2^2$
- Gravity: $D_1 = (m_1+m_2)\ g\ L_1\ c_1 + m_2\ g\ L_2\ c_{12}$, $D_2 = m_2\ g\ L_2\ c_{12}$

Meaning of terms: D_{ii} = effective inertia at joint i, D_{ij} = coupling inertia, D_{iji} = **centripetal** force due to speed of joint j on joint i, D_{ijk} = **Coriolis** force from joints j and k, D_i = gravity load.

9.4 Newton-Euler (N-E) formulation

A **force balance** method: apply Newton's law F = m a and Euler's law N = I alpha + omega x I omega to each link.

- It is **recursive**: a forward recursion (base to tip) finds velocities and accelerations, then a backward recursion (tip to base) finds forces and torques.

Steps of N-E method

- Step 2: **Outward recursion** $i = 1$ to n: $omega_i$, $alpha_i$, linear acceleration of link i and its centre of mass.
- Step 3: Compute the inertial force $F_i = m_i\ a_{ci}$ and moment $N_i = I_i\ alpha_i + omega_i\ x\ (\ I_i\ omega_i\ )$.
- Step 4: **Inward recursion** $i = n$ to 1: $f_i = F_i + f_{i+1}$, and $n_i = N_i + n_{i+1}$ + position cross-force terms.
- Step 5: Joint torque $tau_i = n_i\ .\ z_{i-1}$ (revolute joint) or force $f_i\ .\ z_{i-1}$

(prismatic joint).

10. Trajectory Planning

10.1 Basic terms

- **Path**: the sequence of positions (and orientations) the hand passes through - a purely geometric description.
- **Trajectory**: the path **with time** - includes velocity and acceleration history.
- **Path planning**: deciding the geometric route, usually through via points between start and goal.
- **Trajectory planning**: computing the time function of each joint (or Cartesian coordinate) so the motion is smooth.
- Requirements: smooth motion, no jerks, joint velocity and acceleration within limits, avoid obstacles.

10.2 Joint space vs Cartesian space planning

Point	Joint space	Cartesian space
What is planned	Joint angles vs time	Hand position vs time
Computation	Light, no inverse kinematics online	Heavy, inverse kinematics at each step
Hand path	Unpredictable curve	Controlled (e.g. straight line)
Singularity	Not a problem	Can cause trouble
Use	Pick and place, fast moves	Welding, gluing, machining

10.3 Joint space - cubic polynomial

Conditions: start angle theta_0 at $t = 0$, end angle theta_f at $t = t_f$, start and end velocities = 0.

$$\text{theta}(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

- Boundary conditions: $\text{theta}(0) = \text{theta}_0$, $\text{theta}(t_f) = \text{theta}_f$, velocity(0)

= 0, velocity(t_f) = 0.

- Solution: a_0 = theta$_0$, a_1 = 0, a_2 = 3 (theta$_f$ - theta$_0$) / t_f^2, a_3 = -2 (theta$_f$ - theta$_0$) / t_f^3
- Velocity is parabolic, acceleration is **linear** (but not continuous at the end points).

Worked example (cubic)

A joint moves from 15 degrees to 75 degrees in 3 seconds, starting and ending at rest. Find the coefficients and the angle at t = 1 s.

- a_0 = 15, a_1 = 0
- a_2 = 3(75 - 15)/9 = 3(60)/9 = **20 deg/s^2**
- a_3 = -2(60)/27 = **-4.44 deg/s^3**
- theta(1) = 15 + 20(1) - 4.44(1) = **30.56 degrees**

The **quintic (5th order)** polynomial is used when acceleration at the ends must also be specified, giving smoother start and stop.

10.6 Cartesian-space trajectories

- **Straight line motion** (point-to-point in Cartesian space): the hand follows a straight line in space between the two end poses.
  - Divide the line into small equal steps. At every step compute the pose, then use **inverse kinematics** to get joint angles.
  - Position: $p(t) = p_0 + s(t)\,(p_f - p_0)$, where s goes from 0 to 1 smoothly.
  - Orientation is interpolated at the same time. Used in welding, sealing, cutting.
  - Cost: many inverse kinematics solutions and the path may pass through a singularity.
- **Skew motion**: the hand moves along a straight line **while its orientation also changes (rotates) at the same time**. It is also used in "slant" straight-line travel where the tool axis is skew to the path. The orientation change is interpolated about a fixed axis by an angle that

goes 0 ⟶ total angle.

- **Joint integrated (joint-interpolated) motion**: all joints start and stop **together**. The joint that has to travel the farthest sets the time, all other joints move proportionally slower. The hand path is a curve (not straight) but the motion is simple and quick.

Worked example (straight line)

Move the hand of a 2-link arm from (1, 1) to (1, 0.5) in 5 equal steps along a straight line.

- Step size in y = (0.5 - 1)/5 = -0.1 m, x constant.
- Points: (1, 1), (1, 0.9), (1, 0.8), (1, 0.7), (1, 0.6), (1, 0.5).
- For each point, solve inverse kinematics for $theta_1$, $theta_2$ and send to the controllers.

11. Obstacle Avoidance and Path Planning

11.2 Configuration space (C-space)

- A point in C-space = one complete set of joint values (q_1, ... q_n).
- Obstacles in workspace are mapped to **C-obstacles** (forbidden regions) in C-space.
- The planning task becomes: find a path for a **single point** from start to goal in the free C-space.

11.3 Common methods

- **Potential field**: goal attracts the arm, obstacles repel it; the arm slides down the combined potential.
  - Simple and fast, but can get trapped in **local minima**.
- **Via-point method (practical)**: user teaches extra via points so that the arm goes around the obstacle.

Key point: obstacle avoidance is easiest in **C-space**, and the final path is checked for collision of **every link**, not only the end-effector.

12. Quick Revision and Important Questions

Formula sheet

- Rot(z, theta): c -s / s c in the x-y block. Inverse of rotation = transpose.
- A-matrix = Rot(z,theta) Trans(0,0,d) Trans(a,0,0) Rot(x,alpha).
- Planar 2R: $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$.
- $\cos(\text{theta}_2) = (x^2 + y^2 - L_1^2 - L_2^2) / (2\ L_1\ L_2)$.
- Jacobian: $V = J\ \dot{q}$. Torque: $\tau = J^T F$. Singular if $\det(J) = 0$.
- Cubic: $a_2 = 3\ (\text{theta}_f - \text{theta}_0) / t_f^2$, $a_3 = -2\ (\text{theta}_f - \text{theta}_0) / t_f^3$.

Important questions

- Derive the forward kinematic equations of a 2-link planar manipulator.
- Derive the inverse kinematic solution for a 2-link arm. Why are there multiple solutions?
- What is the Jacobian? Derive it for a 2R arm and explain singularity.
- Explain Lagrange-Euler formulation. Derive the torque equations for a 2-link arm.
- Explain joint-space and Cartesian-space trajectory planning. Derive the cubic polynomial.

Exam tip: draw the diagram of the arm with link lengths and angles first, then show the D-H table, then the final answer. Neat diagrams earn marks.

UNIT - V
Robot Programming & Applications

Automation and Robotics (23A03606) - handwritten study notes

Syllabus Topics

- Robot Programming: methods of programming, requirements and features of programming languages, software packages, problems with programming languages
- Robot Application in Manufacturing: material transfer, material handling, loading and unloading
- Process applications: spot welding, continuous arc welding, spray painting
- Assembly and Inspection

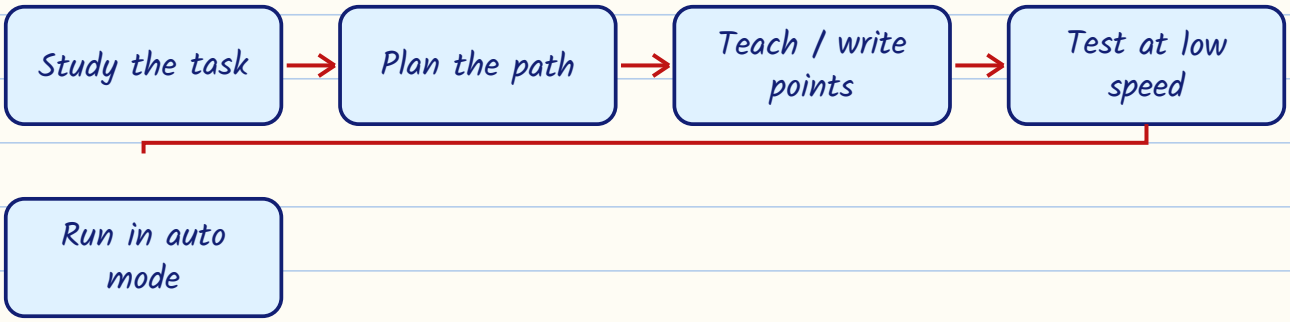
Exam tip: Unit V is mostly descriptive. Learn the three programming methods, the features of a good robot language, and one clear example of each application with its robot requirements.

1. Robot Programming - Introduction

A robot is useful only when it is **programmed**. Programming means teaching the manipulator the sequence of motions, positions, speeds and I/O actions needed to perform a task.

- A robot program is a list of instructions the controller executes in order
- It contains **motion commands** (go to a point), **position data** (coordinates, joint values), **sensor / I/O commands** (wait for signal) and **program-flow commands** (IF, GOTO, loops)
- The controller stores the program in memory and replays it, so the robot is called a **teach and playback** machine

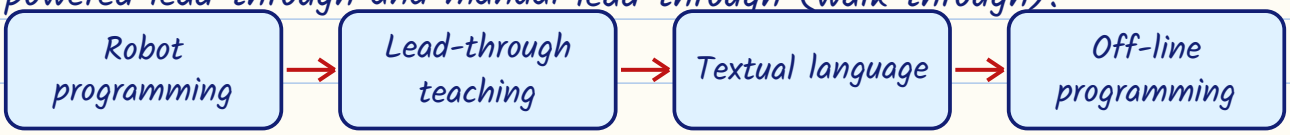
Programming workflow



A robot program = positions (where) + motions (how) + logic and I/O (when). All three must be defined.

2. Methods of Robot Programming

There are three broad approaches: **lead-through**, **textual (computer-like) languages** and **off-line programming**. Lead-through is further divided into powered lead-through and manual lead-through (walk-through).



2.1 Lead-through programming

The programmer **physically leads** the robot through the required motion and the controller records the points. During playback the robot repeats the recorded path.

(a) Powered lead-through (teach pendant)

- Uses a hand-held **teach pendant** with buttons / joystick / keys
- Each axis or the end effector is moved by keys in joint, world or tool coordinates
- When the arm reaches the desired position the programmer presses **RECORD / TEACH** and the point is stored
- Speed, function codes (open gripper, weld on) are added for each point
- Mostly used for **point-to-point** (PTP) tasks such as spot welding, pick and place

(b) Manual lead-through (walk-through)

- The programmer **grasps the arm** (or a dummy / light teach handle) and moves it manually through the complete path
- Controller samples the joint positions at a fixed time interval and stores them
- Suited for **continuous-path** tasks: spray painting, arc welding, polishing
- Robot must be light, or a **teach arm** (scaled replica) is used for heavy robots

Comparison of lead-through methods

Feature	Powered lead-through	Manual lead-through
Teaching device	Teach pendant	Operator moves the arm by hand
Path type	Point to point	Continuous path
Points stored	Selected key points	Many sampled points
Typical use	Spot welding, loading	Painting, seam welding
Accuracy of points	High	Depends on operator skill
Memory needed	Small	Large

Advantages of lead-through

- Easy to learn, no knowledge of a computer language is needed
- Programmer works with the real robot, so positions are exact for the real cell
- Quick for simple tasks

Disadvantages of lead-through

- Robot is **out of production** while being taught
- Difficult to add logic, branching, calculations and sensor decisions
- Hard to edit long programs and not easy to reuse for a similar part
- Danger to the programmer who works inside the work envelope

Teach mode safety: reduced speed (max 250 mm/s), enabling (dead-man) switch and emergency stop on the pendant.

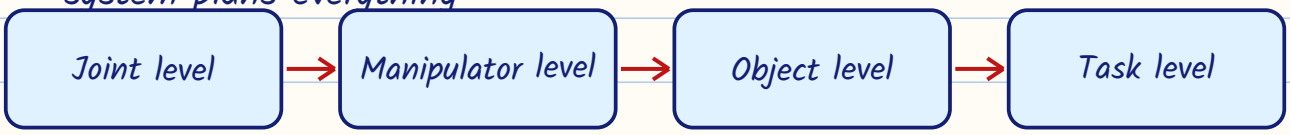
3. Textual (Computer-like) Robot Languages

In **textual programming** the programmer writes statements in a robot language using a keyboard and editor. The program is compiled or interpreted by the controller.

- Positions are defined as variables, e.g. frames and points, either taught once or calculated
- Supports **branching, loops, subroutines, arithmetic, sensor input and communication** with other machines
- Also called **explicit programming** or **robot-level programming**

Levels of robot programming languages

- **Joint level**: program the individual joint values
- **Manipulator level**: program the end effector position in Cartesian space (VAL, AL)
- **Object level**: command is on the objects (e.g. PLACE PART ON TABLE) and the system plans the motion
- **Task level**: only the final goal is given (e.g. ASSEMBLE GEARBOX); the system plans everything



> Higher level = easier for the user but needs a more powerful controller with world model and planner.

4. Some Popular Languages

Language	Developer	Remarks
VAL / VAL II	Unimation (PUMA)	Interpreter based, easy, widely taught
AL	Stanford University	Research, concurrent motions, force control
AML	IBM	Manufacturing language for IBM robots
RAPID	ABB	Modern structured language, very readable

KRL	KUKA	Used for KUKA robots
KAREL	FANUC	Pascal-like structured language
URScript	Universal Robots	Python-like for cobots

VAL style example (pick and place)

A short program in VAL-like syntax is shown below, written as plain lines:

- PROGRAM PICKPLACE
  - SPEED 50
  - MOVE P1 (go to home)
  - APPRO P2, 50 (approach 50 mm above part)
  - MOVE P2 (go down to part)
  - CLOSEI (close gripper)
  - DEPART 50 (lift part)
  - MOVE P3 (go to place location)
  - OPENI (release part)
  - END

5. Requirements and Features of Robot Programming Languages

Requirements of a good robot language

- **Easy to learn and use**, readable, English-like commands
- Should be able to define **positions, frames and transformations**
- Should handle **motion control**: speed, acceleration, interpolation and path types
- Should support **sensor data** and decision making
- Should support **real-time** response and interrupts
- Should allow **editing, debugging and testing** without stopping production
- Must provide **safety** features (limits, collision handling, error recovery)

Features of robot programming languages

- **Motion statements**: point-to-point, linear, circular, spline interpolation
- **Data types**: integer, real, boolean, string, joint, location (frame), vector,

array

- **Program flow control**: IF-THEN-ELSE, FOR, WHILE, CASE, GOTO, subroutine call
- **Input / output handling**: digital and analogue I/O, wait-for-signal, interrupts
- **Sensor integration**: vision, force, proximity, tactile
- **Communication**: serial, Ethernet, fieldbus, with PLC and other robots
- **Multi-tasking** (parallel programs) and **error handling**
- **Frames and offsets**: work frame, tool frame, relative moves
- **Simulation and debugging tools**: breakpoints, single step, trace

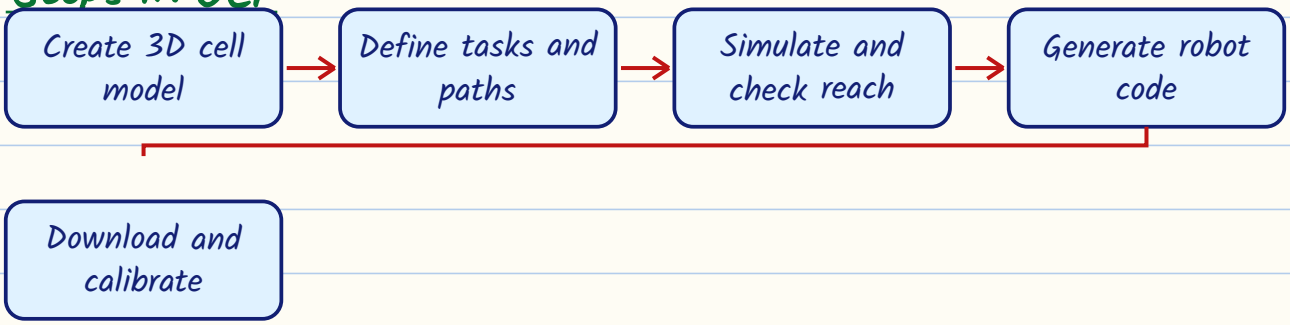
Position = Frame x Offset

A language must let the user write the program once in terms of frames, so that the same program works if the fixture is moved - only the frame is re-taught.

6. Off-Line Programming (OLP) and Software Packages

In **off-line programming** the program is prepared on a computer using a **3D model** of the robot and the cell. The robot is not needed, so production continues.

Steps in OLP



Advantages of off-line programming

- No downtime of the real robot - programming and production in parallel

- Safe: collisions, reach problems and cycle time are checked in the simulation
- Best layout can be chosen before buying equipment
- Programs are stored, copied, edited and reused easily
- Complex paths (welding seams, 3D curves) are generated from CAD data
- Easier integration with CAD / CAM

Disadvantages

- The model never matches the real cell exactly, so **calibration** is required
- Needs trained programmers and costly software
- Errors in the CAD model are carried into the program

Common software packages

Package	Company / use
RoboGuide	FANUC - simulation and OLP
RobotStudio	ABB - simulation and RAPID code generation
KUKA.Sim	KUKA robots
RoboDK	Many robot brands, CAD based
Process Simulate	Siemens - manufacturing cell simulation
DELMIA / IGRIP	Dassault - workcell design
ROS + Gazebo / MoveIt	Open source robot software and simulation

OLP needs calibration, lead-through does not. This is the main trade-off.

7. Problems with Robot Programming Languages

- **No standard** language: every manufacturer has its own, so programs are not portable
- Programmers must learn a new language for every brand
- Limited **sensor and vision** support in many languages
- Difficult to program **complex tasks** such as assembly with force

feedback

- Real-time and multitasking facilities are weak in older languages
- Debugging is poor because the robot moves in the real world, mistakes can cause damage
- Hard to describe 3D geometry in text, hence need for graphic OLP
- **Inaccuracy** of the robot model (link flexibility, backlash, thermal expansion) causes position errors
- Languages do not easily support **error recovery** and exception handling
- Need for trained personnel increases cost

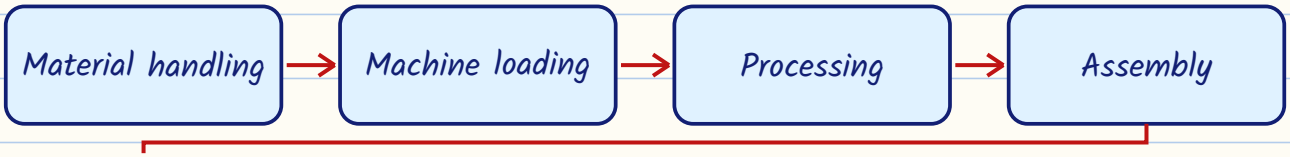
Likely 5 mark question: list the problems of robot languages and suggest solutions.

8. Robot Applications in Manufacturing - Overview

Industrial robots are used where tasks are **repetitive**, **hazardous**, **heavy** or **need high accuracy**.

Main application groups

Group	Examples
Material transfer	Pick and place, palletising, packaging
Machine loading / unloading	CNC lathe, press, die casting, injection moulding
Processing operations	Spot welding, arc welding, spray painting, grinding, drilling
Assembly	Insertion, screwing, fitting, PCB assembly
Inspection	Gauging, vision inspection, testing



Inspection

9. Material Transfer and Material Handling

Material handling: movement, storage, control and protection of materials during manufacture. Robots do the **transfer** of parts from one place to another.

Types of material transfer applications

- **Pick and place**: pick part from one location, place at another (simple 2-4 DOF robots)
- **Palletising and depalletising**: stacking boxes in layers using a defined pattern
- **Packaging**: placing products into cartons
- **Transfer between machines**: along a production line
- **Conveyor tracking**: robot picks parts from a moving conveyor
- **Sorting** with vision

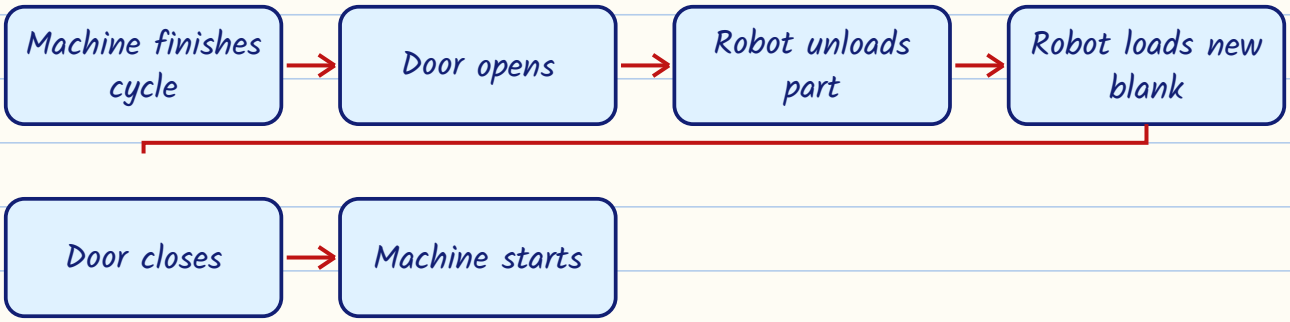
10. Machine Loading and Unloading

A robot **feeds raw parts into a machine** (loading) and **removes finished parts** (unloading).

Applications

- **Machine tools**: CNC lathes and milling machines
- **Press working**: load blanks into presses, remove stamped parts
- **Die casting and forging**: handle hot parts, spray the die
- **Plastic injection moulding**: remove moulded part and runners
- **Heat treatment** furnaces

Robot-machine interface



- Handshake signals: machine READY, door OPEN / CLOSED, part PRESENT, cycle COMPLETE
- Robot waits in a safe position until the signal is received (WAIT instruction)
- A **double gripper** loads a new part and removes the finished one in a single entry

In machine loading the robot is a **slave** of the machine: the machine controls the cycle and the robot responds to its signals.

11. Spot Welding

Resistance spot welding: two sheets are clamped between copper electrodes and heated by a large current for a short time so that they fuse at one spot.

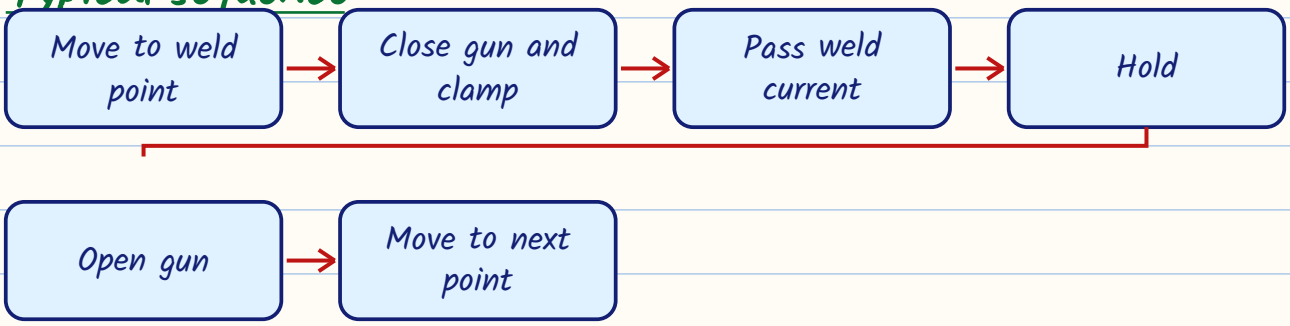
Weld parameters

- Current (5 to 20 kA), weld time (cycles), electrode force and hold time

Robot requirements for spot welding

- **Point-to-point** control with **high repeatability** (about 0.5 mm)
- Payload: weld gun weighs 30 to 100 kg, so a heavy payload articulated robot (6 axes) is used
- Large work volume and **wrist flexibility** to reach the joints in a car body
- Programmed with the **teach pendant**: weld points are taught one by one
- Weld controller interface: start weld, weld complete, gun open / close

Typical sequence



Advantages

- Consistent weld quality, high speed, no operator fatigue or burns

Limitations

- Accessibility of the joint to the gun, heavy weld gun reduces speed

Spot welding = point to point. Arc welding = continuous path. Remember this pair.

12. Continuous Arc Welding

In **arc welding** an electric arc between an electrode (wire) and the workpiece melts the metal, joining the two parts along a **seam**.

Common processes

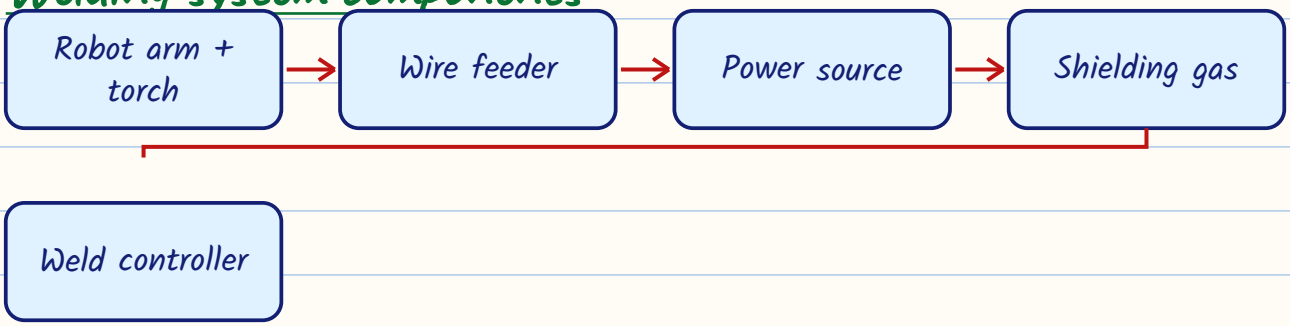
- **MIG / MAG (GMAW)**: continuous wire feed with shielding gas - most common with robots
- **TIG (GTAW)**: tungsten electrode, filler rod, high quality for thin material

Robot requirements

- **Continuous-path** control with straight, circular and weave motions
- Accurate control of **travel speed**, constant torch angle and stand-off distance
- Repeatability around 0.1 mm, payload about 5 to 15 kg
- 6 axes for orientation freedom, often with a **positioner** (2 or 3 axis table) to rotate the work
- Programming: by teach pendant using a few taught points, or manual

lead-through, or OLP from CAD

Welding system components



Advantages

- Uniform weld quality and good penetration, higher arc-on time, safe for the worker (fumes, UV)

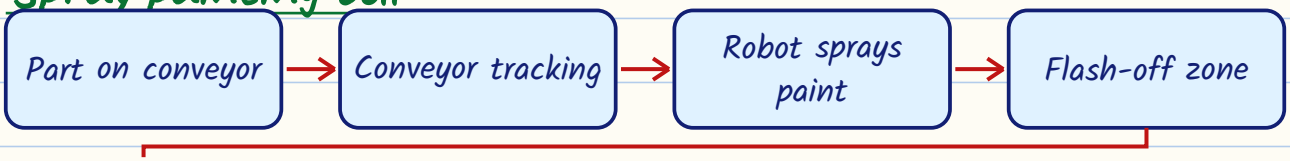
13. Spray Painting

Robots are widely used for **spray painting and coating** because paint fumes are toxic and flammable and quality has to be uniform.

Features and requirements

- **Continuous-path** control with smooth motion and high speed
- Programmed by **manual lead-through** (walk-through): an expert painter moves the arm and the path is recorded and played back
- Explosion-proof design: **hydraulic or intrinsically safe electric** drive, sealed arm, pressurised cabinets
- Large work volume and **many DOF** (6 to 7 axes) to coat complex shapes
- Memory capacity high due to the large number of stored path points
- Control of paint flow, atomising air, gun trigger and colour change

Spray painting cell



Oven curing

Advantages

- Uniform coat thickness, less paint waste (saves 20 to 30 percent paint)
- Operator protected from fumes and noise
- Fast colour change, consistent finish
- Reduced ventilation costs

Painting robots avoid electric sparks: explosion-proof enclosure with positive air pressure is mandatory.

14. Assembly

Assembly is about 40 to 50 percent of the manufacturing cost, so automation here is valuable.

Robots for assembly

- **SCARA robots**: high speed, compliant in the horizontal plane, ideal for vertical insertion (PCB, small parts)
- **Cartesian robots**: accurate and rigid, for simple assembly
- **Articulated robots** and dual-arm robots: complex assembly

Assembly tasks

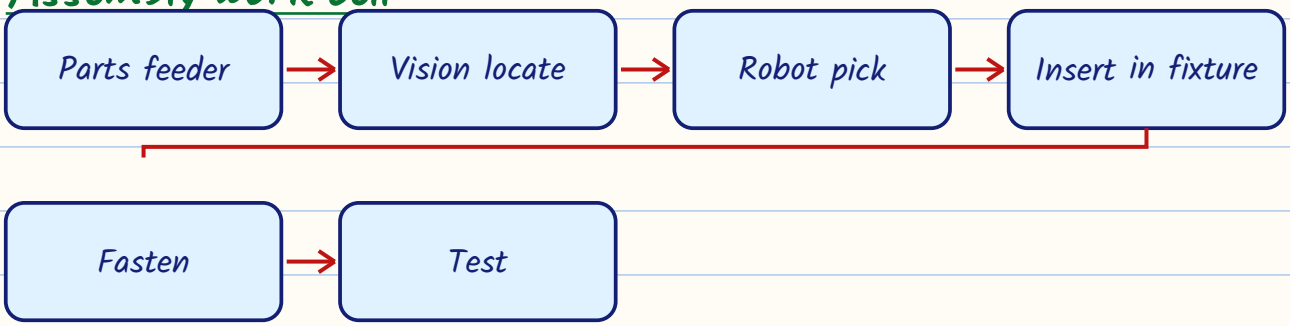
- Part insertion (peg in hole), screw fastening, snap fit
- Adhesive application, riveting, press fit
- PCB component insertion (electronic assembly)

Requirements

- **High repeatability** (0.01 to 0.05 mm) and moderate speed
- Compliance for insertion: **RCC (Remote Centre Compliance)** device to correct small errors
- Force / torque sensing for fitting parts
- Vision to find part orientation, fixtures and feeders to supply parts

- Compact, light grippers, quick tool changers

Assembly work cell



> SCARA = Selective Compliance Assembly Robot Arm - the standard assembly robot.

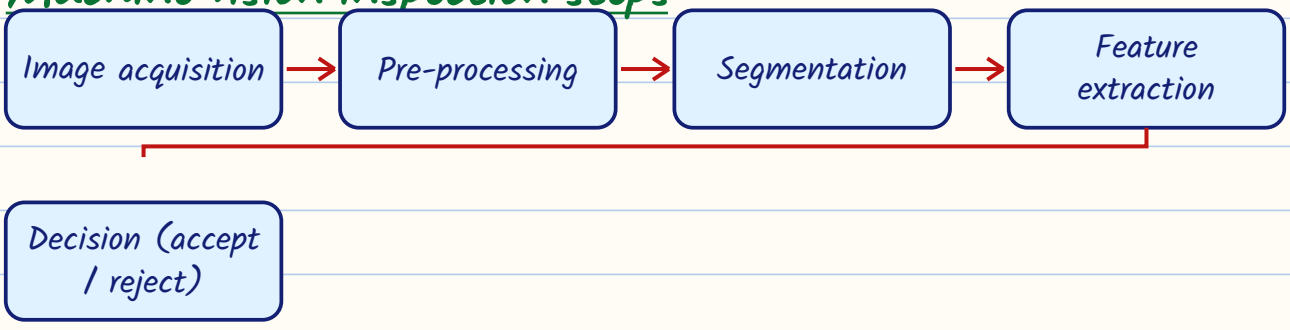
15. Inspection

Robot inspection checks parts for **dimensions**, **defects**, **presence of components** and **surface quality** automatically.

Inspection methods

- **Contact inspection**: touch probe or gauge mounted on the robot - used like a CMM
- **Non-contact inspection**: vision camera, laser scanner, ultrasonic, eddy current, X-ray

Machine vision inspection steps



Advantages

- 100 percent inspection at line speed, objective and repeatable results
- Data recorded for **statistical process control** (SPC)

- Safe in hazardous places such as nuclear plants

Limitations

- Lighting and part presentation affect results
- Robot accuracy is less than a dedicated CMM
- Calibration needed

16. Summary and Quick Revision

Programming

Topic	Key points
Lead-through	Powered (pendant, PTP), manual (walk-through, CP)
Textual languages	VAL, AML, AL, RAPID, KRL; branching, sensors, subroutines
Off-line	3D simulation, no downtime, needs calibration
Problems	No standard, poor sensor support, debugging risk

Important exam questions

- Explain the methods of robot programming with neat sketches
- Compare powered lead-through and manual lead-through
- List the requirements and features of a robot programming language
- Write a VAL program for a pick and place task
- Describe off-line programming and its advantages and disadvantages
- What are the problems with robot programming languages?
- Discuss the robot requirements for spot welding and arc welding
- Explain the use of robots in spray painting
- Explain robot machine loading and unloading with an example
- Write short notes on robot assembly and inspection

Last minute tip: for every application write (1) task, (2) type of control PTP or CP, (3) robot type and features, (4) advantages. This scores full marks.